



# UNIVERSITÀ DEGLI STUDI DI PADOVA

Dipartimento di Fisica e Astronomia “Galileo Galilei”

Master Degree in Astrophysics and Cosmology

Final Dissertation

## Binary Stars Beyond the Main Sequence and the Enigma of UV-Dim Stars

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Academic Year 2024/25

## Abstract

Binary stars play a crucial role in the dynamics and evolution of star clusters. However, our understanding of their frequency and distribution beyond the Main Sequence (MS) remains limited. Another intriguing yet enigmatic phenomenon involves UV-dim stars, whose presence in very young clusters has not been fully explored, leaving open the question of whether this trait emerges at cluster formation or develops over time. In this thesis, I derived high-precision multi-band photometry obtained from Hubble Space Telescope images of the young, star-forming cluster NGC 346 and the intermediate-age cluster BS 90, both located in the Small Magellanic Cloud and within the same field of view. Using this photometry, I investigate two distinct, unrelated topics: the occurrence of binary systems in evolved stellar stages, and the presence of UV-dim stars in young environments.

To constrain the binary fraction in the Horizontal Branch (HB) and Red Giant Branch (RGB), I have combined the derived photometry with synthetic binary modeling, enabling the first direct identification and quantification of binary systems beyond the MS. I applied this method to three clusters of different ages: the old Galactic globular cluster 47 Tucanae, the intermediate-age cluster BS 90, and the young Large Magellanic Cloud cluster NGC 1783, developing a new approach for estimating binary fractions in evolved stellar stages, and extending the investigation to environments where the MS is not easily accessible. The method accounts for contamination by field stars and asymptotic giant branch (AGB) stars, refining the results based on the quality of the data for each cluster. The findings reveal significant differences in binary fraction between the MS, HB, and RGB. For instance, in 47 Tucanae, I derived a binary fraction of  $f_{bin}^{q>0.9} = 0.040 \pm 0.012$  in the HB and  $f_{bin}^{q>0.9} = 0.020 \pm 0.004$  in the RGB, compared to a literature value of  $f_{bin}^{q>0.5} = 0.009 \pm 0.003$  for the MS. Assuming a flat mass-ratio distribution, the inferred total binary fraction from HB and RGB stars appears inconsistent with the one derived from the MS. This paradox suggests two possible explanations: either the mass-ratio distribution is peaked around unity, making the binary fractions compatible, or the observed discrepancy arises from the presence of Blue Straggler Stars (BSSs), which could influence the binary fraction estimates in evolved populations. Following this second scenario, I derived the BSS fraction in the RGB for 47 Tucanae of  $f_{BSS} = 0.018 \pm 0.004$ . Finally, I investigate the presence of UV-dim stars in the very young cluster NGC 346. The detection of UV-dim candidates in this  $\sim 3$  Myr-old cluster, for the first time, provides evidence that this phenomenon may be an intrinsic feature of early cluster evolution. The presence of UV-dim stars in such a young system challenges existing theories about their origin, suggesting that they may form much earlier than previously thought. This discovery opens new paths for understanding their formation mechanisms and the role of circumstellar dust in shaping their observed properties.

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# Chapter 1

## Introduction

Stellar clusters are the basic building blocks of galaxies and provide fundamental information about star formation, stellar evolution, and galaxy assembly. From ancient globular clusters (GCs) to young open clusters and star-forming complexes, these systems are laboratories for probing the physical processes that govern stellar populations. The study of clusters provides insights into how stars form in different environments, how they evolve dynamically over time, and how they form galaxies.

Although they may appear as simple gravitationally bound systems of stars, star clusters are not. In fact, most clusters contain multiple populations of stars with a variety of different chemical compositions and evolutionary histories, contrary to the classical view of their evolution as simple stellar populations. In addition, young and dense high-mass clusters have a profound effect on the surrounding interstellar medium, while globular clusters give us a glimpse into the distant past and the evolution of galaxies. Therefore, studies of cluster formation and evolution are of paramount importance for a wide range of astrophysical research. Their dynamical interactions influence the evolution of binary systems, the formation of black holes, and the presence of unusual stellar objects such as Blue Stragglers (BSSs). By observing clusters of different ages and environments, we are able to gain a better understanding of stellar physics, galactic evolution, and the overall structure formation in the Universe.

Among all the different phenomena that characterize clusters, this thesis will focus on two of them: binary stars and UV-dim stars. In particular, the investigation will focus on how the binary fraction varies outside the Main Sequence (MS), focusing on three clusters with different ages, and on the presence of UV-dim stars in a very young cluster in the Small Magellanic Cloud, namely NGC 346.

## 1.1 Binary Stars in Clusters

Binary stars play a significant role in the cluster dynamical evolution, serving as a source of heating and providing crucial insights into the interpretation of stellar populations in clusters. Through the study of the binaries frequencies, it is possible to shed light on cluster dynamics and evolution. Binary systems are also thought to be the progenitors of an ensemble of exotic objects, such as Blue Stragglers (BSSs), milli-second pulsars, cataclysmic variables and X-ray binaries. Therefore the determination of the fraction of binaries plays a fundamental role towards the understanding of the origin and evolution of these peculiar objects.

### 1.1.1 Detection of Binaries

There are three main methods for measuring the fraction of binaries in star clusters (Hut et al. 1992). The first relies on identifying binaries through radial velocity variability measurements and needs the detection of each binary system individually. The second method is based on the search for photometric variables: this technique needs photometric surveys on wide fields, therefore is time consuming because of the necessity to repeat measure in different epochs.

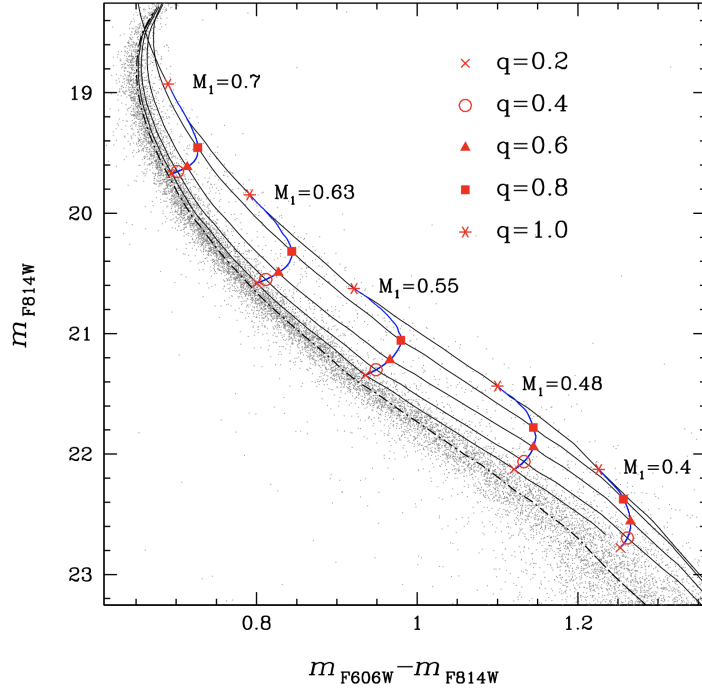
The third method is the most efficient for measuring the fraction of binaries in a cluster. It is based on the analysis of the number of stars located on the red side of the main sequence fiducial line and is the optimal method for several reasons:

- it is a statistically robust method due to the availability of a large number of stars;
- it is efficient in terms of observational time since two filters are enough for detecting binaries and there is no need for repeated measures;
- it is sensitive to binaries with any orbital period and inclination.

This approach was used by Milone et al. (2012) to study the binary populations in individual stellar clusters.

To be more specific, this method relies on the fact that binaries that are able to survive in the dense environment of a GC are so close that even the Hubble Space Telescope is not able to resolve the single components; therefore, the light coming from each star will combine and the binary system will appear as a single point-like source, brighter than a single star of the same mass.

If we consider the two components of binary system and we indicate with  $m_1$ ,  $m_2$ ,  $F_1$



**Figure 1.1:** Model MS-MS binary sequences with different mass ratios. The dashed-dotted line is the MSRL while continuous black lines indicates the sequences of constant  $q$  and blue lines are the sequences of constant  $\mathcal{M}_1$  (Milone et al. 2012).

and  $F_2$  their magnitude and fluxes, the binary will appear as a single object with magnitude:

$$m_{\text{bin}} = m_1 - 2.5 \log \left( 1 + \frac{F_1}{F_2} \right) \quad (1.1)$$

If the binary is formed by two MS stars the fluxes are related to the stellar masses ( $\mathcal{M}_1, \mathcal{M}_2$ ) and its luminosity will depend on the mass ratio  $q = \frac{\mathcal{M}_1}{\mathcal{M}_2}$  (we assume  $\mathcal{M}_2 < \mathcal{M}_1, q < 1$ ). In the case of equal-mass binaries, they will then form a sequence almost parallel to the MS, around 0.75 mag brighter. If the masses of the two components are different, the binary will appear redder and brighter than the primary and populate a CMD region on the red side of the MS ridge line (MSRL), which represents the locus of single MS stars in the CMD, but below the equal-mass binary line, based on their  $q$ , as can be seen in the CMD in Fig.,1.1. Therefore, the fraction of stars that lie in between the MSRL and the equal-mass binary line is the key to determine the binary fraction of the GC. However, it is important to note that binaries with a small mass ratio are not very far from the MSRL, making them hard to distinguish from single MS stars. For this reason, it is impossible to directly measure the general population of binaries without assuming a specific distribution of mass ratios  $f(q)$ .

The main challenges of this method are that it requires high-quality photometry and

high-resolution imaging to minimize the fraction of blended stars; it can be affected by differential reddening, which spreads out the MS and makes it more difficult to distinguish the binary sequence from the single-star MS population. Lastly, it requires an accurate analysis of photometric errors as well as a correct estimate of the field contamination to distinguish real binaries from bad photometry and field objects.

### 1.1.2 Strengths and Limitations of Previous Studies

The previous studies of binaries in the MS (e.g. Milone et al. 2012, Mohandasan et al. 2024, Cordoni et al. 2023) have revolutionized our understanding of the binary fraction and binary properties in globular clusters. The MS is very suitable for such studies because binaries are more easily identifiable there from their position in CMDs. However, such studies are also not without limitations. In distant or poorly populated clusters, MS detection can be troublesome due to photometric depth restrictions or crowding, and it cannot be subjected to complete binary analysis. Also, the partial knowledge of the mass ratio distribution remains a basic restriction to the complete description of the binary population in star clusters. To go beyond these limitations, it is important to extend the study of binaries beyond the MS and into sequences that remain visible even in more extreme environments. Although sequences such as the Horizontal Branch (HB) and the Red Giant Branch (RGB) do not provide the same advantages as the MS, such as a wide stellar mass spectrum, they can nonetheless provide information about the binary populations in clusters where the MS cannot be resolved.

In this thesis, I will develop and apply a method to study binaries in the HB and RGB across three clusters of different ages: 47 Tucanae, a well-studied old galactic GC that serves as an excellent analog for similar cluster populations outside the Milky Way, an intermediate-age cluster in the Small Magellanic Cloud, BS 90, which provides insight into binary evolution at an earlier stage, and a young cluster, NGC 1783, located in the Large Magellanic Cloud, where binaries can be studied in a recently formed stellar environment. Investigating clusters with different ages allows me to see whether the resulting binary fractions are related to this parameter. This approach will help assess the potentiality of binary studies beyond the MS and expand our understanding of binary populations in a broader range of conditions.

## 1.2 The Horizontal Branch

The Horizontal Branch (HB) is the stage of stellar evolution that follows the red-giant branch (RGB) in stars with masses similar to that of the Sun. The stars at this stage are supported by helium core fusion (via the triple-alpha process) and hydrogen shell fusion. The helium core fusion ignition on the RGB tip introduces drastic change in the star structure, with a general decrease in the luminosity, some envelope contraction, and surface temperatures rising. Knowledge of the HB is critical for understanding stellar evolution and for the determination of stellar populations of GCs.

The HB stars share the same core mass following the helium flash, as evidenced in their comparable luminosity. It is therefore upon plotting in a CMD diagram by visual magnitude that the branch takes the form of a horizontal line. However, such a simplistic image is not supported by the structure of the HB. In fact, different globular clusters tend to have different HB morphologies.

Stellar evolution studies show that for fixed age and helium abundance, metal-rich stars at the RGB tip are more massive and colder than stars with lower metallicity  $Z$ . Hence, they would reach a lower effective temperature and redder colors than metal-poor stars when approaching the HB. Therefore, metallicity is commonly considered to be the main parameter governing HB morphology in globular clusters: metal-rich GCs host red HBs while metal-poor ones typically host blue HBs. However, it soon became clear that the morphology of the HB could be widely different even in GCs with similar age and metallicity and that a "second parameter" was at play (e.g. Sandage et al. 1967; Fusi Pecci et al. 1993).

In the years, several second parameters have been proposed, including cluster age, helium abundance, mass loss, and stellar rotation. A major obstacle to understanding the observed HB of GCs is that the parameters at play are often degenerate, so that a number of different combinations of age, metallicity, and helium provide similar HBs. The evidence that nearly all GCs host multiple stellar populations with different helium abundances has provided an additional challenge to explain their HBs.

### 1.2.1 Binaries and Horizontal Branch Morphology

The binary fraction has also been suggested as a possible second parameter that influences the shape of the HB. There is some evidence for a correlation (Lei et al. 2014) while other studies find marginal or zero effects (Milone et al. 2012).

Following Lei et al. (2014), the tidally enhanced stellar wind would affect the HB morphology of GCs by enhancing the primary star mass loss in binary evolution. Therefore,

binary is a potential candidate for the second parameter of GCs because binary fraction plays a significant role in the tidally enhanced stellar-wind model. The authors, in their paper, investigated the impact of binary fractions on the morphology of HB, eliminating any potential effect of metallicity and age. As per their result, the observed effect of the binary fraction on HB morphology may be overpowered by the very strong impact of metallicity and age. However, after removing the effects of metallicity and age, one realizes that HB approaches a bluer color with increasing binary fractions. The result above contradicts the procedure adopted by Milone et al. (2012), who determined that there is either zero or a small correlation between HB morphology and binary fraction in their sample observations.

In reality, they did not eliminate the influence of other significant parameters while investigating the influence of binary fraction on HB morphology, e.g., the mentioned metallicity and age. Note that this was not the central theme of their work but an incidental outcome. This discrepancy highlights the need for a more rigorous reevaluation of the relationship between binary fraction and HB morphology, employing advanced methodologies and better datasets.

As a first step towards this purpose, I focus on the analysis of the binary fraction within the horizontal branch regime of certain globular clusters of different ages. However I won't draw ultimate conclusions about its direct effect on HB morphology, as such a consideration requires a more thorough analysis of other parameters and physical mechanisms.

### 1.3 The phenomenon of UV-dim stars

Young and intermediate-age clusters of both Magellanic Clouds exhibit complex color-magnitude diagrams. In addition to the extended main sequence turn-off (eMSTO), observed in clusters younger than  $\sim 2$  Gyr, cluster younger than  $\sim 800$  Myr exhibit split main sequences. These comprise a blue MS, composed of stars with low rotation rates and a red MS, hosting fast rotating stars. It is widely accepted that stellar populations with different rotation rates are responsible for both these phenomena but their formation and evolution is still debated. Milone et al. (2023b) discovered a new feature along the eMSTO of NGC 1783: a group of stars extremely dim in ultraviolet bands, therefore located on the red side of the eMSTO in CMDs composed of UV filters. These UV-dim stars, which comprises around the 7% of the total number of eMSTO stars of NGC 1783, define a narrow sequence and have colors similar to the remaining eMSTO stars when observed in optical CMDs.

In Milone et al. (2023a), they investigated five clusters younger than  $\sim 200$  Myr and concluded that UV-dim stars are common in young cluster and that most of them populate the bright portion of the blue MS, which is composed of slow-rotating stars. D'Antona et al.

(2023) investigated the physical mechanisms responsible for the UV-dim stars, comparing the observed CMDs of NGC 1783 with simulated photometry by assuming that some stars are obscured by dust. Since the amount of absorption due to dust depends on the wavelength, they reproduced the colors of UV-dim stars by assuming that the dust is distributed in a ring and the stars are seen nearly equator-on. They suggested that circumstellar dust is associated with mass-loss events that can preferentially occur in fast-rotating stars. The fact that UV-dim stars in young clusters are slow rotators could be a challenge for this scenario, however they suggested that the blue MSs of young clusters may be composed of stars that were initially fast-rotating but that have slowed down (D’Antona et al. 2017). In this context, Milone et al. (2023a) speculated that UV-dim stars in young clusters are indeed stars that have been recently subject to an intense mass-loss event which caused both their abnormal UV absorption and braked their rotation shifting their location to the slow-rotating MS.

In this thesis, I will discuss the case of UV-dim stars in NGC 346, an extremely young cluster of approximately 3 Myr, located south of BS 90. A study of UV-dim stars within an extremely young cluster is an opportunity to investigate, for the first time, whether this phenomenon is present since the formation time of the cluster or whether it occurs later on in its existence. Understanding their presence at such an early stage may also help in finding a suitable explanation for their nature. Since NGC 346 shares the same field of view as BS 90, for which I performed the data reduction, it serves as an ideal candidate for this study. Its young age makes it particularly relevant, and its location allows me to analyze two distinct and unrelated phenomena while utilizing the same data reduction process.

## 1.4 Thesis Layout

This Chapter was dedicated to a comprehensive presentation of the phenomenon of binaries in star clusters and to UV-dim stars. With the aim to extend the analysis of the binary fraction to other regions of the CMD, as the Horizontal Branch and the Red Giant Branch, in this thesis I use a method similar to the one introduced in Section 1.1.1 to explore these regions for three different clusters with different ages: 47 Tucanae, an old cluster in the Milky Way, BS 90, an intermediate-age cluster located in the SMC and NGC 1783, a young cluster part of the LMC.

In Chapter 2, I outline the procedure followed to reduce the observational data for the clusters NGC 346 and BS 90 in the Small Magellanic Cloud. Chapter 3 covers the

analysis which was performed in order to detect the binary fraction in these clusters, with brief investigation of UV-dim stars in the young star-forming cluster NGC 346. Finally, in Chapter 4, I present the findings, their implications, and potential directions for future work.



## Chapter 2

# Observations and Data Reduction

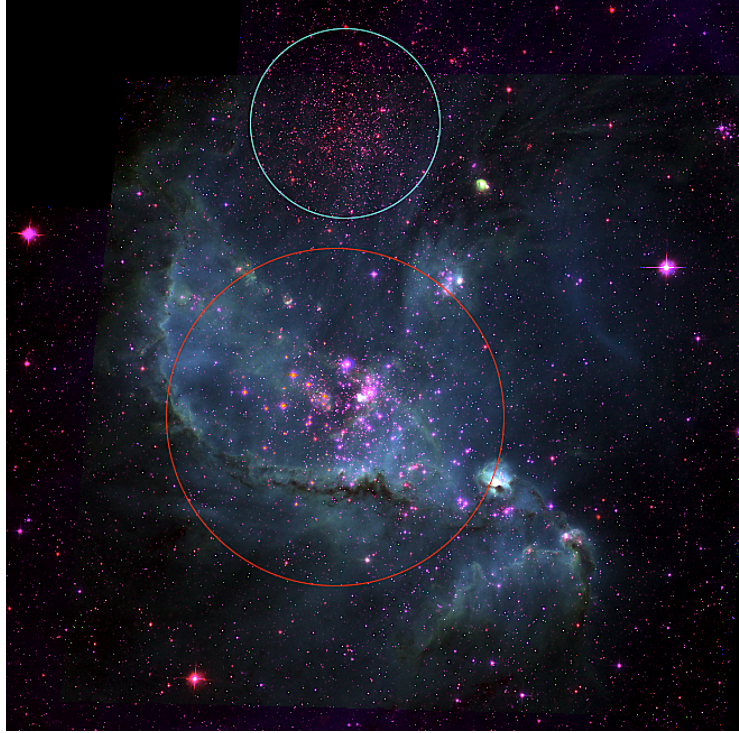
In this Chapter I will illustrate the procedure I followed to obtain the photometric stellar catalog used to derive the CMDs of the young SMC cluster NGC 346 and the intermediate-age star cluster BS 90, visible to the north of NGC 346. An ACS image of NGC 346 and BS 90 is shown in Fig. 2.1 where the two clusters are marked with a red and a blue circle, respectively.

### 2.1 Instrumentation

The photometric data used in this study were obtained by the Hubble Space Telescope (HST), one of the most powerful space-based telescopes that allow high-precision photometry. HST has been orbiting around the Earth since 1990 at an altitude of 540 km, allowing to take high-resolution, distortion-free images in a wide range of wavelengths. The telescope is equipped with a primary mirror measuring 2.4 m in diameter and a focal length of 54.6 m. The telescope is able to observe a broad wavelength range, from near-infrared to ultraviolet. To achieve this, two cameras are required: the Advanced Camera for Surveys (ACS) and the Wide Field Camera 3 (WFC3).

The ACS was designed for wide-field imaging in the visible wavelength range and is composed of three channels: the High-Resolution Channel (HRC), the Solar Blind Channel (SBC) and the Wide Field Channel (WFC). WFC3 is composed by two different channels, one to observe in the UV and optical range (WFC3/UVIS), and one to observe in the IR (WFC3/IR). In this study I used data obtained with the following channels:

- **WFC/UVIS** – It has a field of view of  $160 \times 160$  arcsec<sup>2</sup> and works in the wavelength range 850 – 1700 nm, allowing very high-resolution observations. Among the available filters I used data acquired through the filter F225W (200 – 290 nm), F275W (229 – 312 nm) and F336W (302 – 371 nm).



**Figure 2.1:** ACS three-color image of the SMC field that includes the clusters NGC 346 and BS 90, highlighted by the red and cyan circle, respectively.

- **ACS/WFC** – It has a field of view of  $202 \times 202 \text{ arcsec}^2$  and its spectral range 350 – 1050 nm. From this instrument I used data obtained through the filters F814W (687 – 963 nm), F555W (458 – 621 nm) and F475W (386 – 556 nm).

## 2.2 Dataset

The astro-photometric catalog used to build the CMDs of NGC 346 and B90 was obtained starting from archive images available in the filter mentioned in Sec. 2.1. In particular the data were collected as part of the programs GO-10248 (P.I.: A.Nota) for the F814W and F555W filters, GO-17118 (P.I.: C. Murray) for the F225W, F275W, F475W, and F336W filters. The details of the observations, including the observation date, principal investigator, filter, and camera used, are summarized in Tab. 2.1. Photometry and astrometry are obtained from calibrated, flat-fielded exposures corrected for the effects of the poor charge-transfer efficiency of the detectors (`_flt`).

| Camera    | Filter | Date           | Program | P.I.      |
|-----------|--------|----------------|---------|-----------|
| ACS/WFC   | F814W  | Jul 13-18 2004 | 10248   | A. Nota   |
| ACS/WFC   | F555W  | Jul 13-18 2004 | 10248   | A. Nota   |
| UVIS/WFC3 | F475W  | Aug 6-7 2023   | 17118   | C. Murray |
| UVIS/WFC3 | F336W  | Aug 6-7 2023   | 17118   | C. Murray |
| UVIS/WFC3 | F275W  | Aug 6-7 2023   | 17118   | C. Murray |
| UVIS/WFC3 | F225W  | Aug 6-7 2023   | 17118   | C. Murray |

**Table 2.1:** Details of the observations used in the data analysis, including the observation date, principal investigator, filter, and camera specifications.

## 2.3 Data Reduction

In this section, I focus on images obtained with the F814W, F555W, and F225W filters, as the data from the other filters were already reduced and did not require further processing. The data reduction was performed using several FORTRAN developed by Jay Anderson (Anderson et al. 2008).

Before proceeding with the data reduction, an initial pre-processing step is required. The files retrieved from the HST archive contain images where the two Charge-Coupled Devices (CCDs) overlap, making them unsuitable for direct use. To separate these images, I used the FORTRAN program `flt2wj2`, which aligns and extracts the individual images from each CCD.

### 2.3.1 Effective Point Spread Function

In HST images of globular clusters, most objects appear as relatively isolated point sources, represented by their central coordinates  $(x, y)$  and their total flux  $(f)$ . The aforementioned parameters are determined through the process of fitting the point spread function (PSF) to the pixel array of the image in question. A PSF model represents the distribution of light from a source in the CCD after it has been processed by the telescope. This process of finding the proper model to fit the observational data is challenging in crowded environments such as a globular cluster.

To determine the PSF I used a code developed by Jay Anderson which employs three different methods for measuring stars, each optimized for specific ranges of stellar luminosity and density. The most suitable method for our source and objectives is method I, which is optimal for bright stars. It derives stellar magnitudes and positions by finding and fitting the best available effective PSF model (e.g. Anderson et al. 2000). The quantities are computed separately on each image, and their average is then derived to refine both magnitudes and positions. In contrast, method II and III perform aperture photometry and are best suited

for faint sources and crowded regions.

A key challenge in this process is obtaining the most accurate PSF model for each CCD, as this is essential for deriving precise stellar profiles. However, determining a star’s position requires knowledge of the PSF, while measuring the PSF itself depends on knowing the stars’ positions and fluxes, leading to an inherent degeneracy. To resolve this, I adopted an iterative procedure, following the approach described in Milone et al. (2023b).

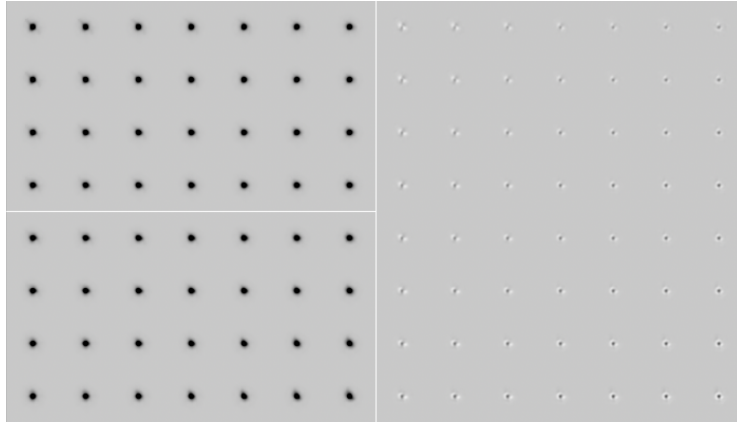
This procedure is implemented using a FORTRAN program called `img2psf` specifically designed for processing images from a selected detector, adapted to reduce images taken with UVIS/WFC3 and WFC/ACS. The main steps of this program are:

1. An initial PSF model is selected from an archival library, based on the CCD and filter properties, providing a first approximation of the PSF.
2. This preliminary PSF is refined using the actual image data to derive a more accurate model.
3. The newly obtained PSF model is then used to measure the position and flux of each source in the image through a fitting process.

This iterative process is applied to bright, non-saturated, and relatively isolated stars to avoid crowded regions and ensure a robust signal for accurately characterizing the PSF.

This routine requires several parameters, including:

- **HMIN** — Defines the isolation index, i.e., the minimum distance between two sources for them to be considered separate. I set this value to 13 px.
- **FMIN** — Specifies the minimum flux a source must have to be included. Since bright sources are required in this initial step, I used a threshold of 4000.
- **PMAX** — Sets the maximum flux count for a source to be considered. This parameter helps exclude saturated stars; I chose 54000, just below the instrument’s saturation limit.
- **QMAX** — Represents the maximum value of the  $Q$  parameter, which quantifies the quality of the fit. It is used to distinguish stars from other objects. I set this to 0.3.
- **NSIDES** — Defines the number of subdivisions for the image, ranging from 1 to 5. This is a trade-off: increasing the number of regions improves the mapping of faint spatial variations in the PSF but reduces the number of stars available for PSF construction. I set  $NSIDES = 2$  to balance these factors.
- **PSFFILE** — Specifies the archival PSF model used to generate the initial guess for the PSF. I used three different PSF models, one for each filter.



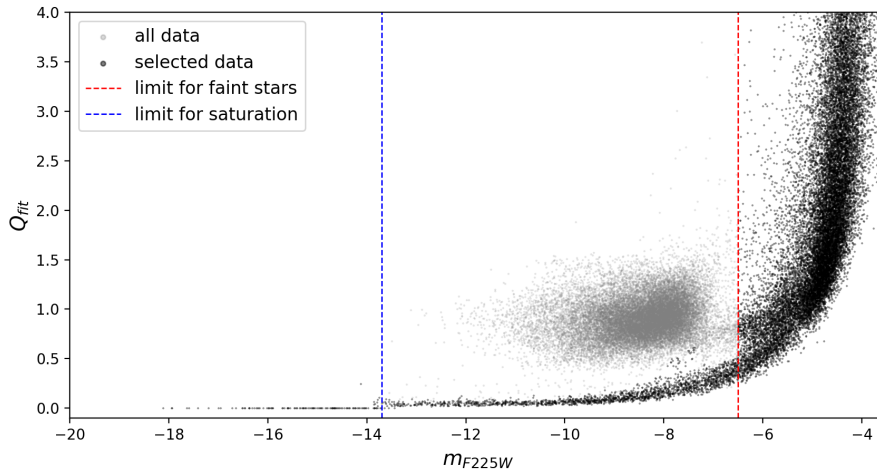
**Figure 2.2:** Example of the grid of PSF models obtained through `img2psf` for an image taken by WFC3 in the filter F225W.

- **IMAGE.fits** — The input image on which PSF measurements are performed.

The program iteratively refines the fitting procedure, using the PSF model from the previous iteration to improve subsequent fits. The process continues until the PSF model stabilizes, meaning that differences between two consecutive iterations become negligible. The final output is a PSF model tailored to the actual image, an example of which is shown in Fig. 2.2.

Rather than generating a single PSF model for the entire image, the program constructs a  $9 \times 10$  grid of PSF models. This approach accounts for spatial variations in the PSF across the CCD, which arise due to optical aberrations and position-dependent charge diffusion. Using a PSF grid enhances the accuracy of position and flux measurements for each source in the image.

The next step involves applying the derived PSF models to the corresponding images to measure the position and magnitude of each detected source. This process is performed using another FORTRAN routine, `img2xym`, which takes as input the PSF models, the image, and several parameters similar to those used in the previous program. Since the PSF now needs to be applied to all detected sources, from faint to saturated stars, the input parameters are less restrictive than before. Specifically, I set  $HMIN = 3$ ,  $FMIN = 50$ , and  $PMAX = 10^9$ . The output of this routine is a file with a `.xym` extension, which contains, for each detected source, the  $(x, y)$  coordinates, instrumental magnitude, and  $Q$  parameter. The  $Q$  parameter, which indicates the quality of the PSF fit and for stellar sources has a well-defined dependence on the magnitude, will be crucial in the following steps to distinguish real stars from other sources.



**Figure 2.3:** ‘Cleaning’ process for one exposure in F225W filter taken as example. The red and blue dashed lines represent the boundaries selected to divide the critical region from faint and saturated stars. The black dots corresponds to accepted stars and the gray cloud are the discarded sources.

### 2.3.2 Cleaning the data

The routine described in the previous section processes all detected sources that meet the selection criteria, treating them as stars. However, some of these sources must be removed, as they may correspond to cosmic rays, hot pixels, PSF artifacts, or noise spikes. To address this, I developed a Python routine that refines the sample based on the  $Q$  parameter. The routine:

- Plots the  $Q$  parameter as a function of instrumental magnitude.
- Defines boundaries to separate the magnitude range into faint stars, critical region (the part of the plot in which the cloud of non-stellar sources is located) and saturated stars.
- Divides the intermediate region into bins.
- Within each bin, selects only the sources with  $Q$  values below a predefined threshold, ensuring that only well-measured stars are retained.

This filtering process improves the reliability of the final catalog by minimizing contamination from non-stellar sources. An example of the output can be seen in Fig. 2.3. This procedure is applied to each image in the F555W, F814W and F225W filters.

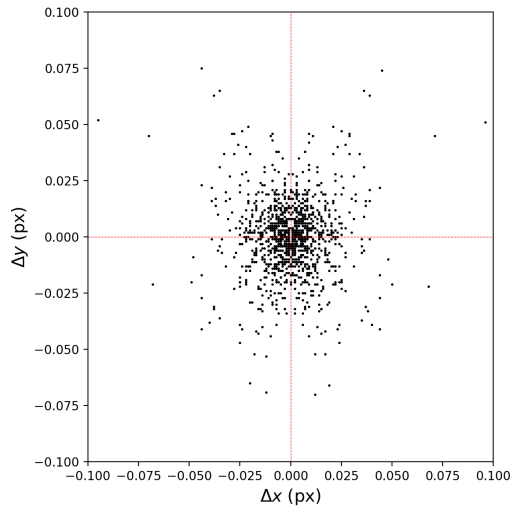
## 2.4 Photometric Catalog

### 2.4.1 Single-filter Catalog

By this stage, one photometric catalog can be constructed for all the stars in BS 90 and NGC 346. To accomplish this, images acquired in each filter must be combined into a single catalog first, which will then serve as the basis of the final photometric catalog. Since the exposures I am combining are not precisely aligned, it is necessary to align them to the same reference frame before combining. To accomplish this, I used the FORTRAN program `xym2mat`: this program computes the 6-parameter transformation matrices needed to align each image to a common reference frame from a given master image, typically the longest exposure time image. The process works by identifying similar triangles composed of stars in the target and reference images and subsequently applying rotations and translations to the triangles until they align. The process, which is known as *cross-identification*, produces an output matrix containing all the necessary transformations.

As input, this program requires a file named `IN.xym2mat`, which specifies the master frame on the first line, followed by a list of all images to be processed. Each entry includes additional codes indicating the camera and filter used for the observations. These codes allow the program to account for CCD- and filter-specific geometric distortions before proceeding with the identification and alignment of similar triangles. The final parameter specified in the input file is the magnitude of the stars used for identifying similar triangles between images. Since I performed two iterations to refine the transformations, in the first iteration, I included all detected stars, including saturated ones. However, in the second iteration, I excluded the saturated stars to enhance the precision of the alignment. In addition to the `IN.xym2mat` file, the program requires an additional input parameter that defines the search radius (in pixels) around each star in the master image, within which it looks for matching stars in the other images. For both iterations, I set this radius to 0.45 pixels. After completing the two iterations, the program produces an output file containing the corrected stellar positions in the master frame for each image.

To verify the success of the procedure, a common test is to plot the residuals provided in the output file. These residuals represent the difference in pixel position between a star in the master frame and the same star in another image after it has been transformed into the master reference frame. If the residuals are distributed in a gaussian-shaped distribution with mean zero and sigma equal to the error on the position, the transformations were successful. Otherwise, if the residuals show systematic trends or large deviations, it indicates that the program did not find the correct transformations, and I should repeat the procedure, either by selecting a different master frame or adjusting the input parameters. An example



**Figure 2.4:** Difference in position after running `xym2mat`, between stars in the master frame and in one of the exposures obtained in the filter F225W brought into the master reference frame. From the distribution of the residuals I can say that this is a successful transformation.

of this test is shown in Fig. 2.4.

Once the correct transformation matrices are obtained, I can construct a stellar catalog for each filter by averaging the positions and magnitudes of all detected sources. To achieve this, I used the FORTRAN program `xym2bar`, which processes the output files from `xym2mat` to compute the average positions and magnitudes of common stars across multiple exposures. As input, this routine requires a file similar to the one used by `xym2mat`, which specifies the stars to be considered. This time, I extend the magnitude range to include and measure all detected stars. Another significant input parameter of this program is `NIMMIN`, defining the minimum number of images in which a star must be detected to be included in the final catalog. I set `NIMMIN = 2` to maximize the number of selected stars. The final output of this process is a file named `MATCHUP.XYEMEE`, which serves as the final catalog for this iteration. This file contains the positions and magnitudes of stars in the master reference frame, along with the RMS values for each quantity, providing an estimate of the measurement uncertainties.

I performed another iteration of `xym2mat`, using as master frame the file `MATCHUP.XYEMEE`, followed by a second iteration of `xym2bar`. The output of this last routine is the final catalog for each filter.



### 2.4.2 Gaia Reference Frame

In the final photometric catalog obtained from the last iteration of `xym2bar`, stellar positions are still referenced to an arbitrary reference frame, namely the one defined by the image that I used as reference. However, since my goal is to build a single, unified catalog, which includes the filters I reduced myself (F225W, F555W, and F814W) and the already reduced data, which are aligned to the Gaia reference frame, I needed to transform the stellar coordinates from the CCD-based frame to the Gaia reference frame. This step ensures that all stars in the catalog share a common astrometric reference, allowing for accurate cross-matching between different filters and external datasets.

To transform the stellar coordinates into the Gaia reference frame, I used one of the already reduced single-filter catalogs, which is already oriented correctly. To link the two catalogs, I employed the FORTRAN routine `xym1mat`, which compares a catalog to a master catalog, identifying common stars and assigning magnitudes in two different filters. This process works similarly to `xym2mat`, relying on the identification of similar triangles. The output is a file with the `.lnk` extension, containing the same number of rows as the master catalog. To obtain the magnitudes for the two filters of each star, along with the Gaia-oriented coordinates, I performed two iterations of `xym1mat`. In the first iteration, I considered only the brightest stars to create a temporary match between the two catalogs, which provided a first guess of the transformation coefficients between the two reference frames. In the second iteration, I considered all stars and used the temporary match from the first iteration. Once I obtained the catalog with the correct orientation, I used the position residuals  $dx$  and  $dy$  as a diagnostic tool to verify the quality of the match, considering only the stars for which  $dr = \sqrt{dx^2 + dy^2} < 0.5$ . Finally, I built three catalogs, one for each filter being processed, where the magnitudes were instrumental, and the positions were referenced to the Gaia reference frame. The final step is to iterate the cycles of `xym2mat` and `xym2bar` with the Gaia-oriented catalog as the master frame. After completing these iterations, I have my three final single-filter catalogs, one for each filter: F225W, F555W, and F814W.

### 2.4.3 Final Catalog

Once the single-filter catalogs are generated, the next step is to link them together to form a unified catalog that contains the positions of the stars and their magnitudes across all the filters. To accomplish this, I once again use the FORTRAN routine `xym1mat`, keeping the F814W catalog as the reference, since it contains the highest number of stars. I match the F814W catalog with each of the other catalogs, obtaining a corresponding `.lnk` file for each filter. Finally, using these `.lnk` files, I construct a catalog that includes all the stars with

Gaia coordinates and HST instrumental magnitudes in the filters F814W, F555W, F225W, F275W, F475W, and F336W.

## 2.5 Calibration

The last step is to calibrate the instrumental magnitudes into the Vega reference system. The result is expressed in terms of calibrated magnitudes  $m_{\text{cal}}$ , using the formula:

$$m_{\text{cal}} = m_{\text{inst}} + \Delta mag + ZP_{\text{filter}} + C \quad (2.1)$$

where  $\Delta mag$  is the difference between the magnitude obtained through PSF photometry and the magnitude obtained through aperture photometry,  $ZP_{\text{filt}}$  is the zero-point in a given filter and  $C$  is the aperture correction.

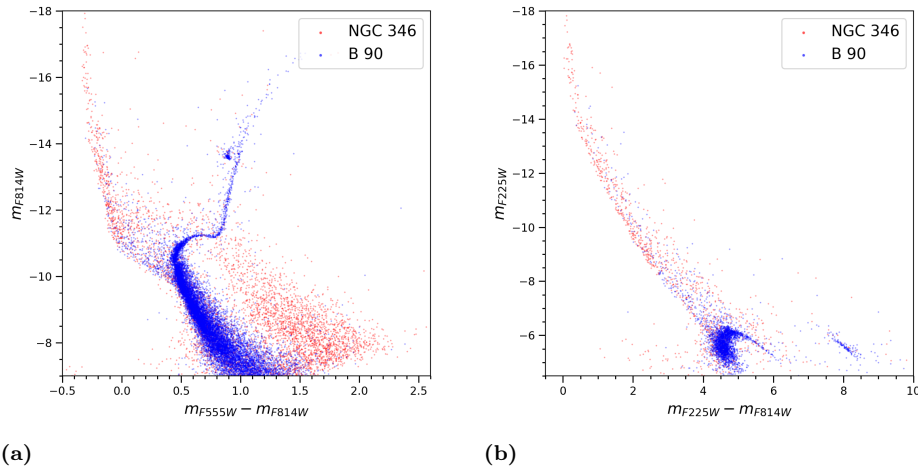
Since the following analysis primarily relies on the relative positions of stars in the CMD, using calibrated magnitudes is not essential for most of the work. The same results can be obtained with instrumental magnitudes. However, calibration is necessary when working with isochrones. For this reason, I calibrated only the magnitudes in the filters required for mass derivation, specifically F814W and F555W.

The reduced images differ from the calibrated ones in two aspects: the calibrated images have an exposure time of one second, whereas the reduced images have longer exposures, and the flux measured in the reduced images is not the total flux of the stars. In order to perform the calibration, I used the `_drc` images, stacked and not suitable for the full data reduction procedure but suitable for the calculation of mean magnitudes rather than for single stellar measurements. They are particularly well-suited for this procedure since they are artifact-cleaned and have a uniform one-second exposure.

On these images, I performed aperture photometry and applied the necessary corrections based on the database, e.g., the aperture correction and the zero-point correction to convert magnitudes to the Vega reference system.

Now I have the calibrated photometry for the stacked images. With the same `xym1mat` program, following the same procedure as before, I aligned my instrumental catalog with my calibrated catalog to identify common stars. The median of the difference between instrumental and calibrated magnitudes provides me with the zero-point correction.

Finally, I applied this zero-point to convert instrumental magnitudes to the Vega magnitude system. As mentioned earlier, only isochrone fitting requires calibrated magnitudes, so all CMDs used in the data analysis prior to that point will be constructed using instrumental



**Figure 2.5:** CMDs with  $m_{F814W}$  vs  $m_{F555W} - m_{F814W}$  (left panel) and  $m_{F225W}$  vs  $m_{F225W} - m_{F814W}$  (right panel).

magnitudes.

## 2.6 Color-Magnitude Diagrams

With the catalog I created, I am now able to plot Color-Magnitude Diagrams (CMDs) to catch a first glimpse of the two Small Magellanic Cloud clusters, NGC 346 and BS 90. In Fig. 2.5 two CMDs are displayed: one using optical filters and the other with the widest wavelength range possible. The figures illustrate the different features of the two clusters, which are primarily attributed to the difference in their age. As mentioned before, NGC 346 is a young cluster while BS 90 is an intermediate-age cluster. Observe that these CMDs are not field-star-cleaned.

The main differences between these two clusters that can be seen from the optical CMD, due to their different ages, are the following:

- **Position of the Main Sequence Turn-Off (MSTO):** In the young cluster, the MSTO is found at a brighter and bluer location than that of the MSTO of the intermediate-age cluster.
- **Horizontal Branch (HB):** The horizontal branch, formed by stars in the helium-core burning phase, is visible in older clusters but may not be present in young clusters, because the stars in the younger cluster have not yet evolved to this phase.

Moreover the fact that for BS 90 the various evolutionary features (MS, SGB, RGB and RC)

are quite tight, show that we are dealing with a simple stellar population (Sabbi et al. [2006](#)).

## Chapter 3

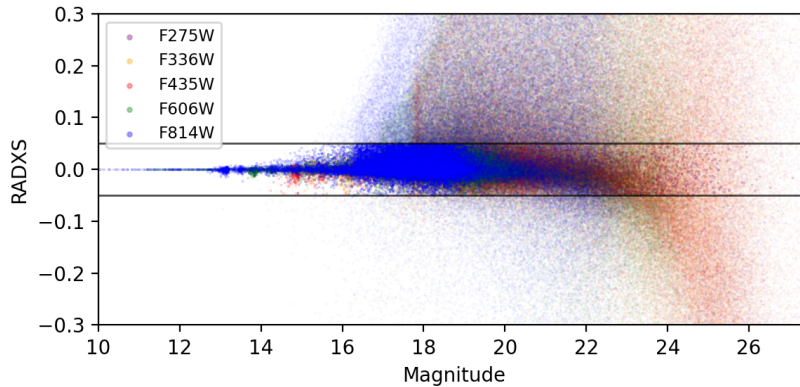
# Data Analysis and Results

This chapter is divided into two parts, as the thesis covers two unrelated phenomena: binaries beyond the MS and UV-dim stars. In the first part, I estimate the binary fraction for various targets of different ages, focusing on distinct regions of the CMD. This analysis will help investigate the presence of binaries in different environments and evolutionary phases. Specifically, I will examine the regions corresponding to the Horizontal Branch (HB) for three clusters: 47 Tucanae (47 Tuc), a relatively old cluster, BS 90, an intermediate-age cluster, and NGC 1783, a young cluster. I will then examine the region of the Red Giant Branch (RGB) only for 47 Tuc and BS 90. In the second part, I analyze the presence of UV-dim stars in the super-young cluster NGC 346.

### 3.1 The Targets

#### 3.1.1 47 Tucanae

47 Tucanae (47 Tuc, NGC 104) is the second-brightest globular cluster in the Milky Way, after Omega Centauri, located near the Small Magellanic Cloud (SMC) in the sky. With an estimated age of 11.8 billion years (Brogaard et al. 2017), 47 Tuc is a perfect example of a dense, ancient stellar system. The cluster is known for its high stellar density, particularly in its core, which increases the likelihood of binary star systems due to frequent close encounters and gravitational interactions between stars. This makes 47 Tuc an ideal environment for studying binary star populations, especially in its dense core, where exotic stars, such as a significant population of blue stragglers, can be found (Parada et al. 2016). 47 Tuc contains at least two distinct stellar populations, offering a unique opportunity to study how binary systems evolve across different phases of stellar evolution. The cluster spans a variety of evolutionary stages, from the main sequence to the horizontal branch (HB) and red giant



**Figure 3.1:** Procedure used to select stars with high-quality photometry, based on the trend of the sharpness parameter,  $RADXS$ , as a function of the magnitude for each filter. The plot illustrates the distribution of  $RADXS$  values against the magnitude, with the horizontal lines indicating the chosen boundaries, specifically  $RADXS = \pm 0.05$ .

branch (RGB), allowing us to investigate how binary interactions influence stars at various points in their lifecycles. Furthermore, 47 Tuc has been extensively observed through both high-resolution imaging and spectroscopic surveys, providing a wealth of observational data.

For this work I'll use the data from the HUGS (HST UV Globular cluster Survey) database (Nardiello et al. 2018, Piotto et al. 2015), retrieved using the photometric method I. The dataset includes stellar position coordinates ( $X, Y$ ), as well as magnitudes in the F275W, F336W, F435W, F606W, and F814W filters. Additionally, it provides the photometric RMS and the sharp parameter ( $RADXS$ ) for each star in each filter, along with a quality fit parameter. The  $RADXS$  is a parameter that indicates the amount of flux that exceeds the predictions from the best-fitting PSF (Bedin et al. 2008), defined as  $RADXS = (\sum_{i,j} \text{pix}_{i,j} - \text{PSF}_{i,j}) / 10^{-\text{mag}/2.5}$  where the sum is calculated within an annulus between 1.0 and 2.5 pixels from the center of the star and is normalized to the star's total flux. It is negative when the object is sharper than the PSF (like in the case of cosmic rays and PSF artifacts) and positive when the object is broader than the PSF (e.g. galaxies). The perfect PSF fit corresponds to  $RADXS = 0$ .

The dataset also contains the membership probability for each star, and its coordinates in right ascension and declination.

To refine the dataset and improve the quality of the resulting CMDs, I applied a selection criterion that filters out less reliable data. Specifically, I selected stars with a membership probability greater than 90% and those with a sharpness parameter  $|RADXS| < 0.05$  in each filter (Fig. 3.1). This filtering approach ensures that only high-quality data, representative of genuine stellar sources, is included in the analysis, helping to produce a cleaner and more accurate CMD for further study.

### 3.1.2 BS 90 and NGC 346

As previously introduced in Chapter 2, these two clusters, NGC 346 and BS 90, are located in the Small Magellanic Cloud (SMC). NGC 346 is a very young cluster (age  $\sim 3$  Myr), while BS 90 is an intermediate-age star cluster formed around 4 Gyr ago (Sabbi et al. 2006). Their extragalactic nature makes them valuable for comparative studies between clusters in our own Milky Way and those in the Magellanic Clouds.

A key advantage of this study is that both clusters lie within the same field of view, allowing me to perform the data reduction process only once while investigating two entirely uncorrelated phenomena. Specifically, I will focus on the intermediate-age cluster, BS 90 to examine the binary fraction, analyzing how the presence of binaries varies across different evolutionary stages. At the same time, NGC 346 will be used to study a separate phenomenon: the presence of UV-dim stars. This approach not only optimizes the data reduction effort but also enables a broader exploration of distinct astrophysical processes occurring in clusters of different ages and conditions within the SMC.

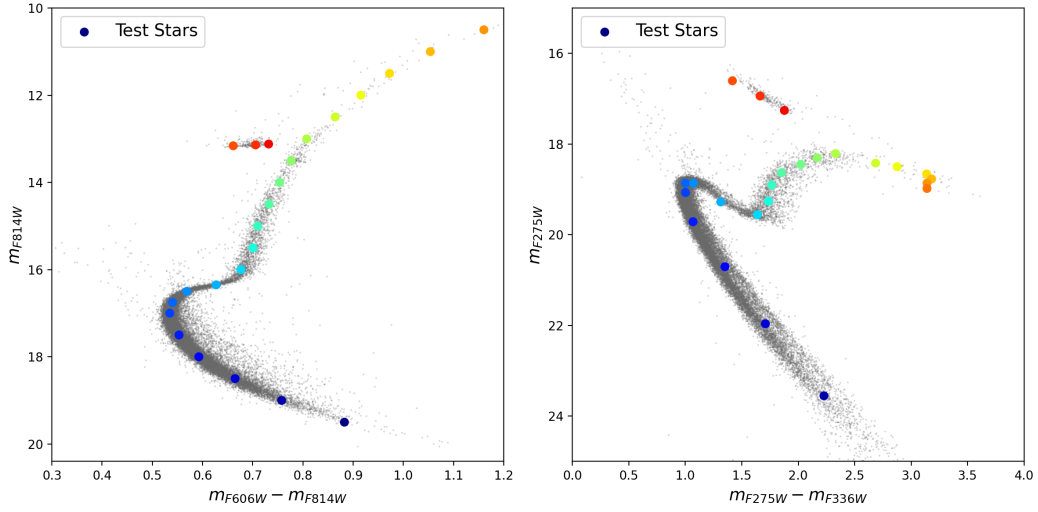
### 3.1.3 NGC 1783

NGC 1783 is a young cluster (1.95 Gyr, Milone et al. 2023b) in the Large Magellanic Cloud (LMC) and is one of the brightest globular clusters in the LMC as seen from Earth. The cluster has been extensively studied in Milone et al. (2023b) as it presents some peculiar characteristics as the upper MS running in a zigzag with two main discontinuities, multiple burst of star formation with an age difference of a few hundred million years, and the presence of a population of UV-dim stars along the extended main-sequence turnoff (eMSTO).

For this study, I will use data from Milone et al. (2023b), which provides stellar position coordinates ( $X, Y$ ), magnitudes in the F275W, F336W, F343N, F435W, F555W, and F814W filters, as well as relative errors and quality flags. The quality flags were used to select stars with the most reliable photometry.

## 3.2 Building Synthetic Binaries for CMD Analysis

Since there are no extensive studies on binary fractions across different evolutionary stages, the first step is to identify the regions of the CMD where binaries are expected to lie. In this analysis, I will focus on binary systems where the primary star belongs to the HB and, separately, on binary systems where the primary star is on the RGB. The same procedure has been applied consistently to both 47 Tuc, BS 90, and NGC 1783. Below, I outline the general workflow.



**Figure 3.2:** Optical and UV CMDs for 47 Tuc, displaying the test stars as filled circles.

### 3.2.1 Defining the Fiducial Line and Selecting Test Stars

To construct the set of test stars for building synthetic binaries, I first selected two fiducial lines: one spanning the entire CMD from the base of the Main Sequence (MS) to the tip of the Red Giant Branch (RGB), and another specifically for the Horizontal Branch (HB). These fiducial lines were initially defined by eye on the  $m_{F814W}$  vs.  $m_{F606W} - m_{F814W}$  CMD. Once the fiducial lines were established, I selected test stars by selecting specific points along these lines corresponding to preselected magnitudes in the F814W filter. Using these points, I calculated the corresponding magnitudes in the F606W filter. To build a complete catalog of test stars, it was necessary to determine the magnitudes of the selected stars in all available filters. To accomplish this, I repeated the fiducial selection over several different CMDs of the form  $m_{F814W}$  vs.  $X - m_{F814W}$ , where  $X$  represents the filter in which I wanted to find the magnitude of each test star.

This procedure was applied to 47 Tuc, as described above, the resulting test stars can be seen in Fig. 3.2. For BS 90 and NGC 1783, I followed the same workflow, with the only difference being the substitution of the F606W filter with F555W, due to the photometric catalog I constructed in Chapter 2 and the one retrieved from Milone et al. (2023b). The resulting test stars for these clusters can be seen in Fig. A.1 and A.2.

The complete catalog of the test stars for 47 Tuc can be found in Tab. B.1, the one for BS 90 in Tab. B.2 and the one for NGC 1783 in Tab. B.3.



### 3.2.2 Building the Binary Systems

Once the test star catalog is built, I can construct the binary systems. This is quite straightforward as we know that if we are looking at a photometric binary, its magnitude will be:

$$m_{bin} = -2.5 \log(F_{bin}) \quad (3.1)$$

where  $F_{bin}$  is the flux of the binary, given by the sum of the fluxes of the single stars,  $F_1 + F_2$ . Therefore knowing the magnitude of the single stars we can derive their flux:

$$F = 10^{-\frac{m}{2.5}} \quad (3.2)$$

and then the flux and the magnitude of the binary system:

$$m_{bin} = -2.5 \log(10^{-\frac{m_1}{2.5}} + 10^{-\frac{m_2}{2.5}}) \quad (3.3)$$

By repeating this procedure for different filters and iterating it over all the test stars, we can determine the position of the binary systems in the CMD. This entire process was carried out using a custom-developed code, which is available on my GitHub repository<sup>1</sup>.

Once the binary systems have been constructed, I can visualize them in a CMD. For each cluster, I chose to use two CMDs: one based on optical filters and the other on UV filters. This allows me to cross-check the results, particularly in the following steps of my analysis. As mentioned earlier, I will focus on binary systems where the primary star belongs to the HB and, separately, on binary systems where the primary star is on the RGB.

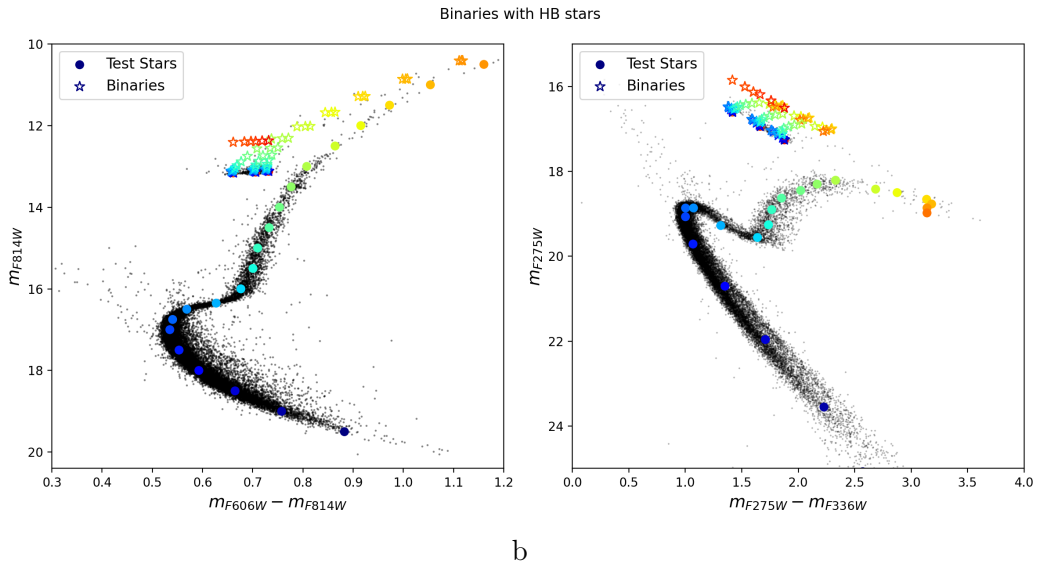
#### Binaries with HB Stars

In Fig. 3.3, I present the first CMDs with the synthetic binaries for 47 Tucanae, focusing on binary systems where the primary star belongs to the HB. In these CMDs, the test stars are represented by filled circles, while the binary systems are shown as empty star-shaped markers. The color of each marker is not arbitrary: it identifies the members of the binary system. Specifically, while the primary star is always on the HB, the secondary star varies. To interpret these CMDs, consider the following example: the blue star-shaped marker represents a binary system consisting of a HB primary star and the secondary star indicated by the blue circular marker.

As shown in the plot, these binary systems generally occupy the region above the HB. Furthermore, if a binary system consists of a HB star and an RGB star, it will be located

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<sup>1</sup>[https://github.com/giadaaggio/master\\_thesis](https://github.com/giadaaggio/master_thesis)



**Figure 3.3:** Optical and UV CMDs for 47 Tuc, displaying the test stars as filled circles and the synthetic binaries with empty star-shaped markers. These CMDs illustrate that binary systems containing a HB star tend to occupy the region above the HB. The red star-shaped markers, in particular, indicates binary systems formed by two HB stars.

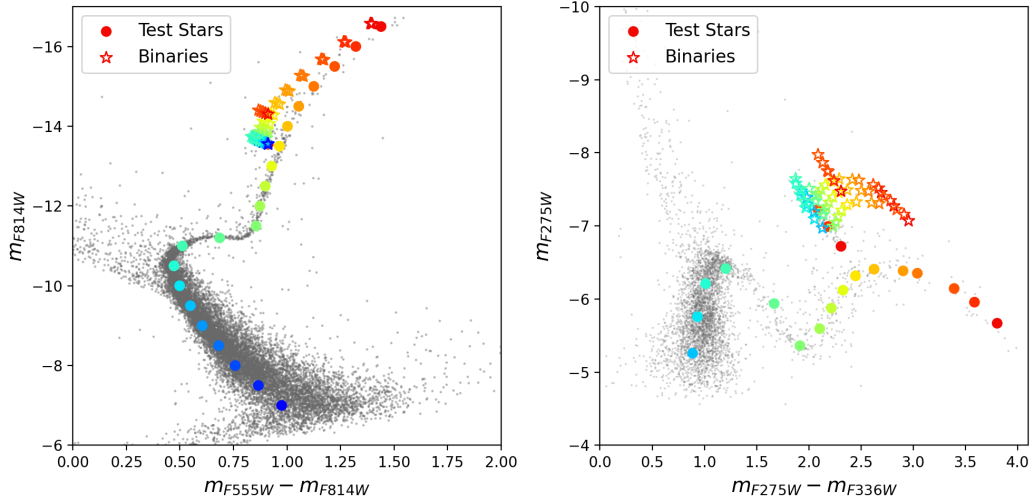
in the Asymptotic Giant Branch (AGB) region in the optical CMD. On the other hand, if the binary is composed of a HB star and a MS star, it will appear closer to the HB in both CMDs, gradually shifting away from it as the secondary star evolves through later stages.

In Fig. 3.4 the same CMDs are shown for BS 90, where the binary systems exhibit a similar distribution, with binaries containing a HB star positioned above the HB region. The same behavior is present in NGC 1783, the relative CMDs can be seen in the Appendix in Fig. A.3.

### Binaries with RGB Stars

In the case of 47 Tuc and BS 90, I repeated the procedure focusing on binary systems where the primary star belongs to the RGB, the results are presented in Fig. 3.5a for 47 Tuc and Fig. 3.5b for BS 90. I was unable to perform this analysis for NGC 1783 due to the small separation between the HB and the RGB.

This time, to improve the readability of the CMDs, I have displayed only the binaries in which the primary star is the same RGB star. The complete CMDs, including all RGB primary stars, can be found in the Appendix (Fig. A.4). Moreover, for these systems, I have only used the optical CMD. In the case of 47 Tuc the RGB in the UV appears broadened due to the presence of multiple stellar populations, while for BS 90 the UV RGB is not well



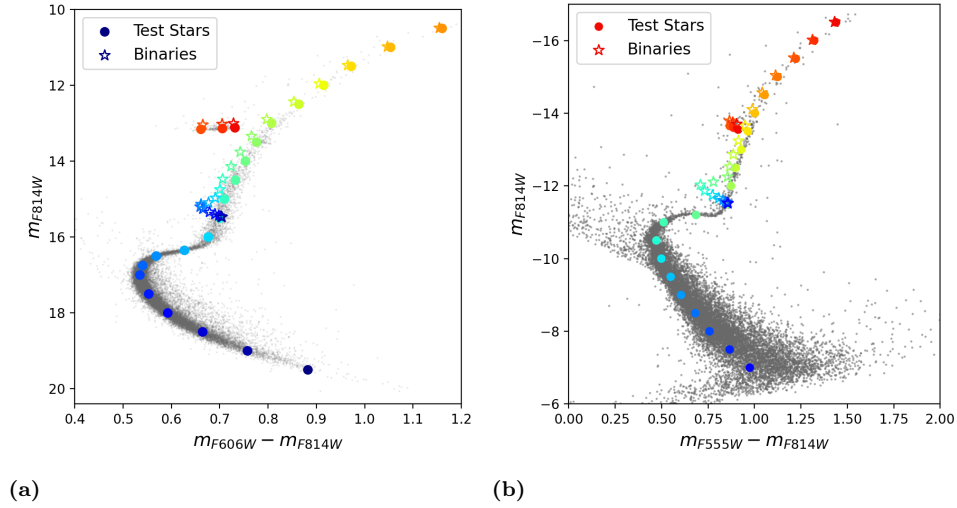
**Figure 3.4:** Optical and UV CMDs for BS 90, displaying the test stars as filled circles and the synthetic binaries with empty star-shaped markers.

defined. This makes it challenging to accurately infer the binary fraction, a topic that will be discussed in the following sections.

As observed in these CMDs, binaries where one member is an RGB star are distributed along a region parallel to the RGB itself, while binaries consisting of an HB star and an RGB star align parallel to the HB. This behavior is consistent with the previous analysis of binaries involving HB stars.

### 3.3 Quantitative Analysis

Once I have established where these binaries are located in the CMD, I proceed with a quantitative analysis to estimate the binary fraction. To this end, I use the method described by Milone et al. (2012), introduced in Chapter 1. This technique involves selecting two regions within the CMD: Region A, in which would lie single stars, and Region B, which must contain binaries based on the distribution of the synthetic systems I have analyzed before. These two regions were selected by eye, with no assumptions, except the visual reference of the distribution of binaries seen in Fig. 3.3, 3.4, and 3.5. The region selection was carried out for both clusters and for both optical and UV CMDs in the quantitative analysis of binaries with one member from the HB. However, for the analysis of the binary fraction with one member from the RGB, the selection was done only for the optical CMD, for the reasons previously introduced.



**Figure 3.5:** Optical CMDs for 47 Tuc (left panel) and for BS 90 (right panel), displaying the test stars as filled circles and the synthetic binaries with empty star-shaped markers. For sake of readability, in these CMDs the binary systems are built using always the same RGB star.

The regions were selected using a custom-made code available on my GitHub repository<sup>2</sup>. An example of the selection of these regions for the region of the HB for 47 Tuc in the optical CMD is presented in Fig. 3.6.

The following step is to count the stars in region A and the ones in region B and then compute the fraction of binaries simply as:

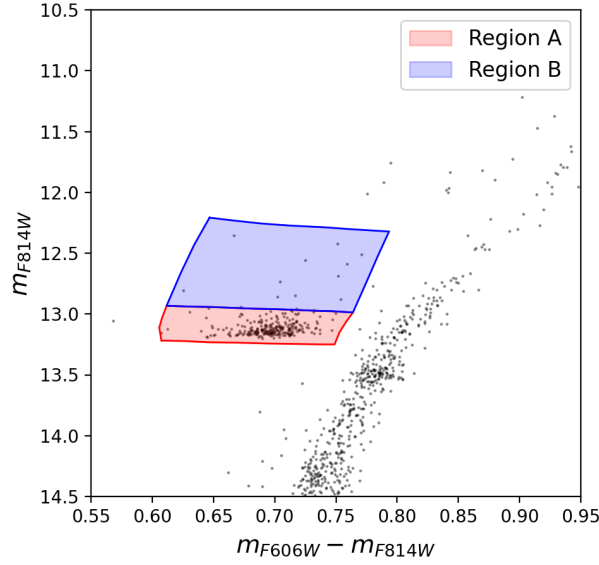
$$f_{\text{bin}} = \frac{N_B}{N_B + N_A} \quad (3.4)$$

where  $N_A$  is the number of stars in the region A and  $N_B$  is the one in region B. The error associated to each quantity of Eq. 3.4 is the Poisson error and the error on the obtained binary fraction is calculated by following the standard error propagation.

To ensure that I distinguish genuine binaries from blends, it is crucial that a binary can be identified in both CMDs. In other words, if a candidate binary does not appear in both the optical and UV CMDs, it is likely a blend rather than a true binary system. To guarantee consistency in selecting regions in both diagrams, I positioned the boundary line between region A and region B to pass through the location of the synthetic binary system constructed from the same pair of stars in both the optical and UV CMDs.

Since I am analyzing clusters in different environments, one in our Galaxy and the other in the Small Magellanic Cloud, I have to use different techniques when counting stars in the areas selected, to avoid counting stars that do not belong to the clusters. In particular, the

<sup>2</sup>[https://github.com/giadaaggio/master\\_thesis](https://github.com/giadaaggio/master_thesis)



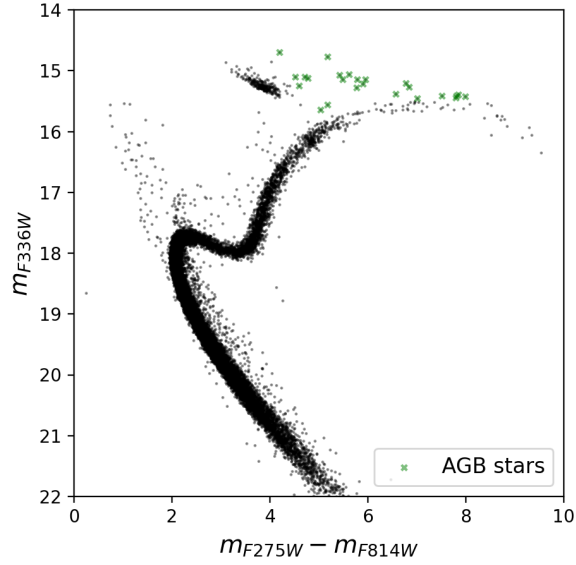
**Figure 3.6:** Example of selection of region A and region B for the HB of 47 Tuc. Red region is region A, in which HB single stars should lie, while region B is colored in blue and should contains binaries.

47 Tuc data are already field-star-cleaned using proper motions, but this was not done for BS 90 and NGC 1783, so a different technique has to be employed to achieve this. Further, in both cases, I will have to take into account the presence of AGB star in the regions selected when computing the binary fraction in the HB. These topics will be discussed later in detail.

### 3.3.1 Binary Fraction in 47 Tucanae

#### Horizontal Branch

The dataset of 47 Tuc does not require additional refining, as it has already been cleaned of field stars using the proper motions. The only caution I must have is to remove AGB stars from the star count within the regions as, specifically in the optical CMD, binaries and AGB stars tend to blend. To achieve this I followed the procedure described in Lagioia et al. (2021), using a far-UV-optical CMD, namely  $m_{F336W}$  vs  $m_{F275W} - m_{F814W}$ . In this kind of CMD, in fact, AGB stars distribute in a specific region, due to the fact that the UV flux of AGB stars is significantly higher than that of RGB stars at similar luminosities. In Fig. 3.7 can be seen the AGB stars that I selected. When counting the stars within the regions A and B to compute the binary fraction, stars marked as AGB were excluded from the count, ensuring a cleaner result.



**Figure 3.7:** Selection of AGB stars in a far-UV-optical CMD. In this kind of CMDs, AGB stars tend to occupy a specific locus in the diagram, making the selection easier.

Finall, I obtained the following values for the binary fraction in the HB region:

$$f_{\text{bin, HB}}^{\text{optical}} = 0.040 \pm 0.012$$

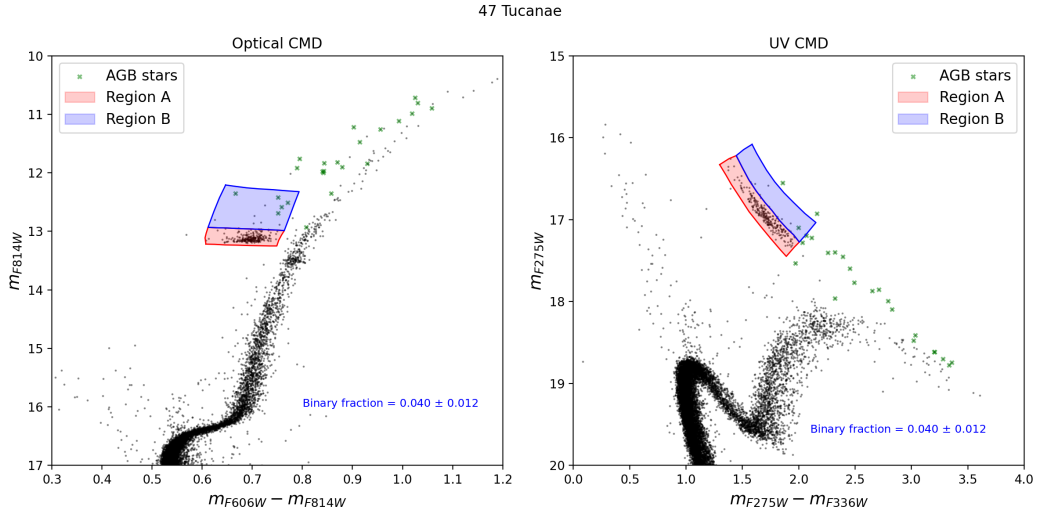
$$f_{\text{bin, HB}}^{\text{UV}} = 0.040 \pm 0.012$$

The selected regions and the AGB stars in the optical and UV CMDs are visible in Fig. 3.8.

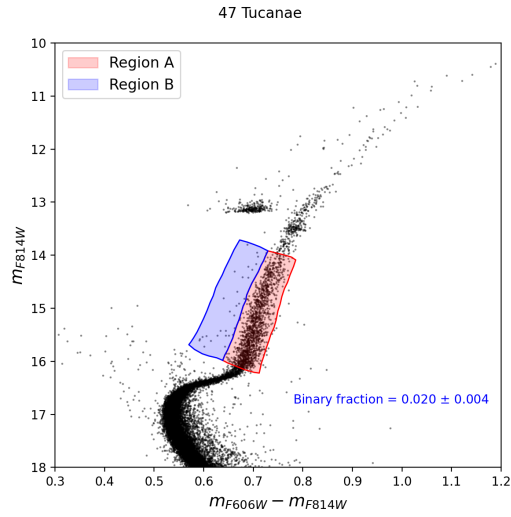
### Red Giant Branch

In computing the binary fraction in the RGB region, I didn't consider particular phenomena since the region where the binaries tend to distribute should be not contaminated by stars in other evolutionary stages. The only limitation in this case is the fact that I cannot use the UV CMD due to the broadening of the sequence caused by the presence of multipopulations. Therefore, I simply selected the regions (Fig. 3.9), counted the stars within them and computed the binary fraction, which resulted in:

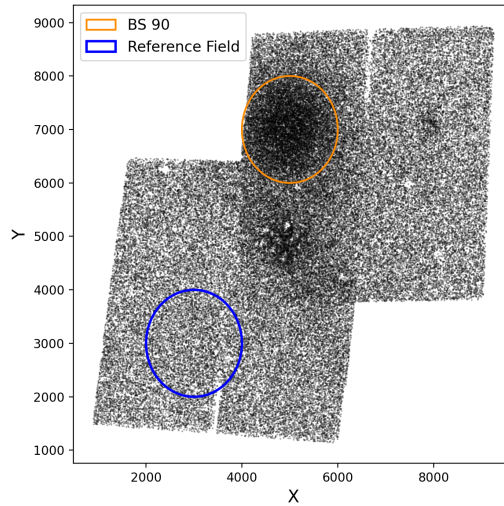
$$f_{\text{bin, RGB}} = 0.020 \pm 0.004$$



**Figure 3.8:** Binary fraction in the HB for 47 Tuc, selected regions and AGB stars in the  $m_{F814W}$  vs  $m_{F606W} - m_{F814W}$  and in the  $m_{F275W}$  vs  $m_{F275W} - m_{F336W}$  CMDs. The green crosses represent AGB stars.



**Figure 3.9:** Binary fraction in the RGB and selected regions in the  $m_{F814W}$  vs  $m_{F606W} - m_{F814W}$  CMD for 47 Tuc.



**Figure 3.10:** BS 90 and the reference field used to account for field contamination. The two circles have the same radius.

### 3.3.2 Binaries in BS 90

For BS 90, which is located in the SMC, field contamination cannot be quantified using stellar proper motions. Therefore, I adopted an alternative approach: selecting a region of the field of view without clusters (Fig. 3.10), constructing CMDs for that region, and counting the stars that fall within the same regions used in the BS 90 CMDs to estimate the binary fraction. In this way I have a first estimate of field contamination. This method has been applied in both the HB and RGB regions.

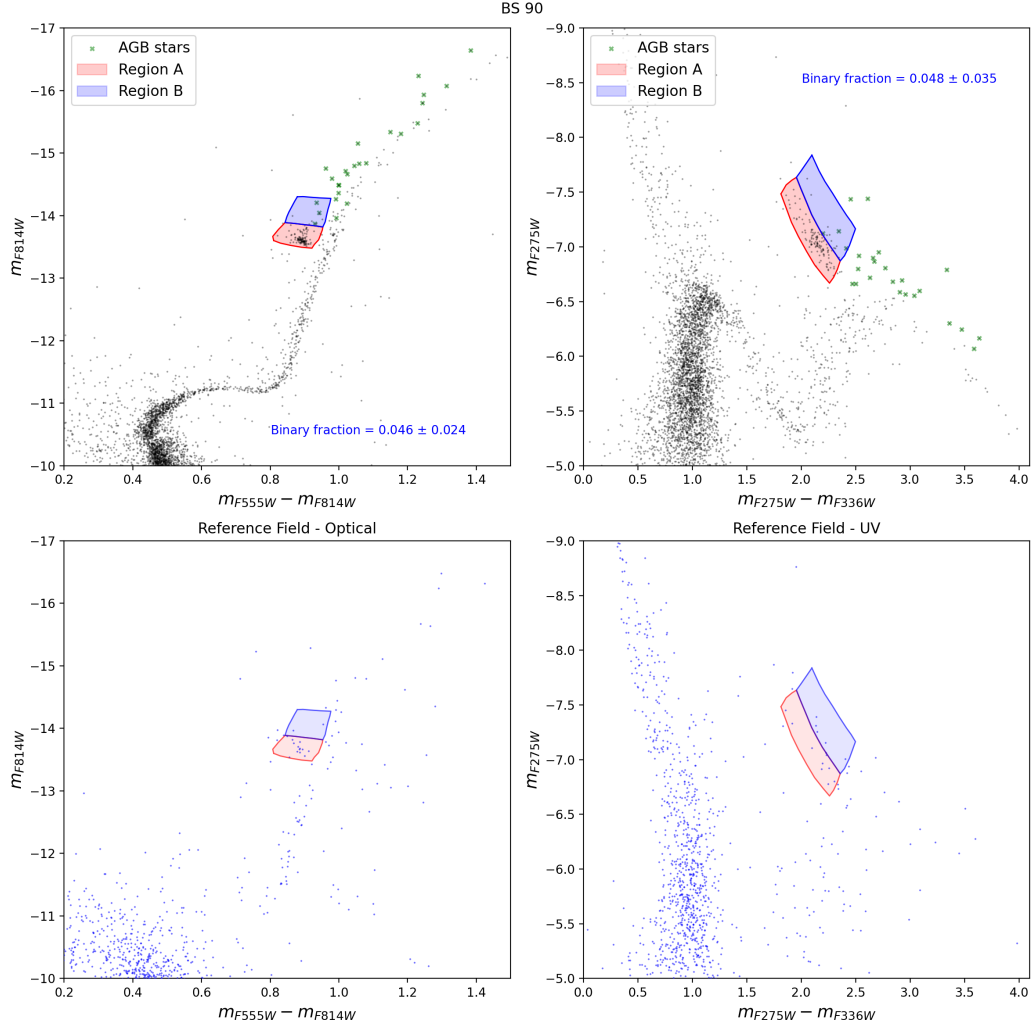
#### Horizontal Branch

In this case as well, I must consider the contamination from the AGB stars. To account for it I proceeded as for 47 Tuc so I selected AGB stars from the far-UV-optical CMD and then I excluded them from the count when computing the number of stars within region A and region B. Fig. 3.11 shows the results of the entire procedure, including region selection and cleaning for AGB and field stars, along with the CMDs of the reference field. In this cluster, in the Horizontal Branch, the binary fraction resulted to be:

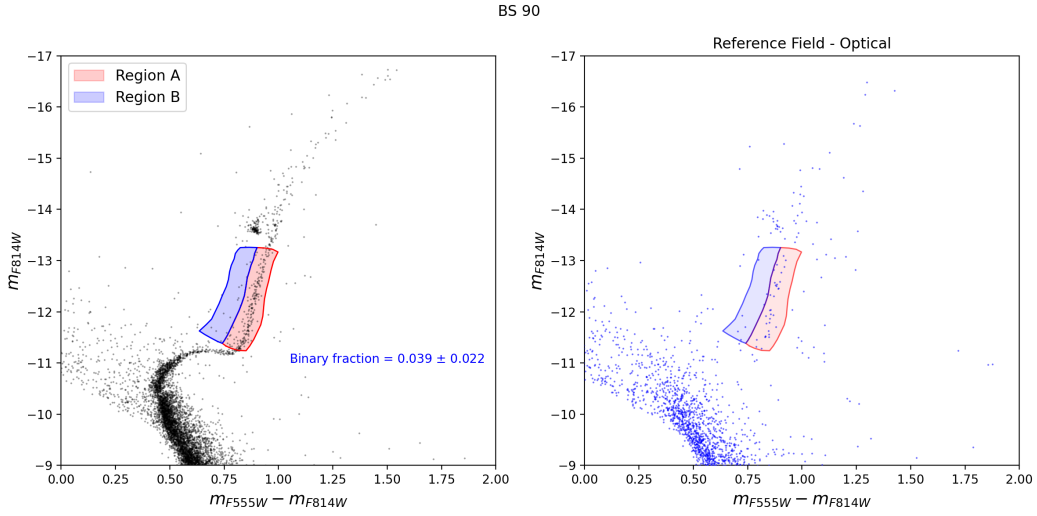
$$f_{\text{bin, HB}}^{\text{optical}} = 0.046 \pm 0.024$$

$$f_{\text{bin, HB}}^{\text{UV}} = 0.048 \pm 0.035$$





**Figure 3.11:** Upper panels: binary fraction in the HB for BS 90, selected regions and AGB stars in the  $m_{F814W}$  vs  $m_{F555W} - m_{F814W}$  and in the  $m_{F275W}$  vs  $m_{F275W} - m_{F336W}$  CMDs. The green crosses represent AGB stars. Lower panels: CMDs for the stars of the reference field used to account for field contamination when computing the binary fractions, the blue and red regions are the same for which I computed the binary fractions.



**Figure 3.12:** Left panel:  $m_{F814W}$  vs  $m_{F555W} - m_{F814W}$  for BS 90. In red and blue are drawn region A and region B, respectively. Right panel: stars of the reference field along with region A and region B.

### Red Giant Branch

Not having to deal with AGB star, the computation of the binary fraction in the RGB for BS 90 is straightforward and followed the same workflow as the one for 47 Tuc. The only difference is the field contamination, for which I proceeded as before. The result is a binary fraction of:

$$f_{\text{bin, RGB}} = 0.039 \pm 0.022$$

As for 47 Tuc, even in this case I made use only of the optical CMD since in the UV CMD of BS 90 the RGB is not clearly defined, making it impossible to select the regions smoothly. In Fig. 3.12 I present the CMD along with the region used to compute the binary fraction and the reference field.

### 3.3.3 Binaries in NGC 1783

For NGC 1783, I applied the same procedure as for BS 90 to account for both field star contamination and the presence of AGB stars. As mentioned earlier, the binary analysis was conducted only for the HB, as the small separation between the RGB and HB made it difficult to analyze the RGB binaries. The final result for the binary fraction in the HB for NGC 1783 is:

$$f_{\text{bin, HB}}^{\text{optical}} = 0.093 \pm 0.021$$

$$f_{\text{bin, HB}}^{\text{UV}} = 0.094 \pm 0.020$$

The CMDs relative to this analysis can be seen in Fig. A.5.

### 3.4 Identifying the Mass Ratio

To compare the binary fractions obtained in this study with those inferred for the Main Sequence, I need to analyze the mass ratio of the binary systems, as it is a key parameter for deriving the total binary fraction in the cluster. As shown in Milone et al. (2012), the total flux of a binary system is determined by the fluxes of its individual stars, which in turn depend on their stellar masses. Consequently, the position of a binary system in the CMD is directly linked to its mass ratio, defined as  $q = \mathcal{M}_2/\mathcal{M}_1$ , where I assume  $\mathcal{M}_2 < \mathcal{M}_1$  and thus  $q < 1$ . As can be seen in Fig. 1.1 for MS-MS binary systems, the binaries with a lower mass ratio are located near the MS while the equal-mass binaries form a sequence almost parallel to the MS. Therefore binaries with large mass ratios have a larger offset in luminosity from the MS and are relatively easy to detect while a small mass ratio doesn't pull them very far off of the MS, making them hard to distinguish from single MS stars. In practice, the limited photometric precision and accuracy makes impossible the direct measure of the overall population of binaries without assuming a specific distribution of mass ratios  $f(q)$ . The same applies to both the HB and the RGB: the binary fractions I determined earlier do not represent the total binary fractions of the clusters. To obtain the total fraction, I need to determine the lowest detectable mass ratio  $q$  and then use a specific mass ratio distribution  $f(q)$  to compute the global binary fraction.

However, while Milone et al. (2012) estimated the fraction of MS-MS binaries by assuming a minimum mass ratio and considering only binaries with  $q > 0.5$ , my approach takes the opposite direction: I first determined the binary fraction and now infer the mass ratio based on the boundaries I defined when selecting regions A and B.

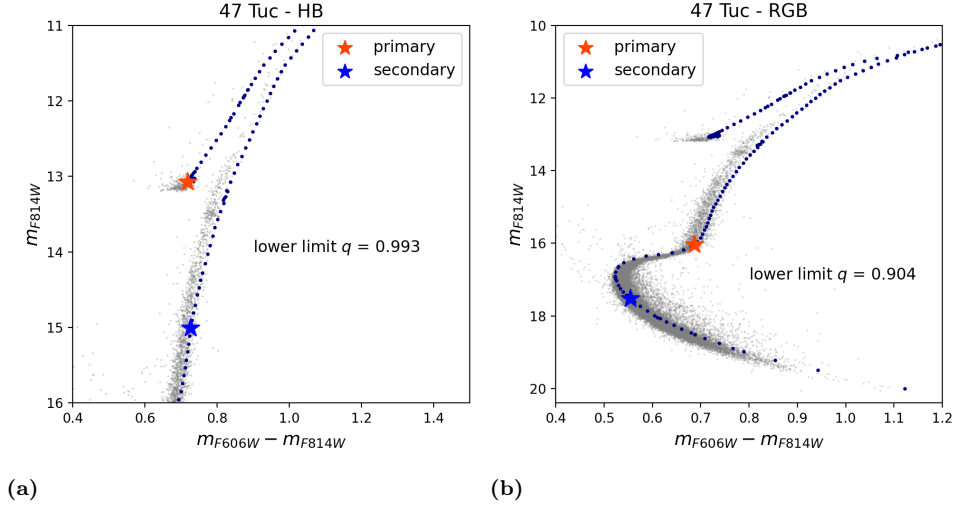
The workflow is the following, applied to both the HB and RGB for all the three clusters that I analyzed.

#### Isochrones Fitting

Since determining mass ratios requires the stellar masses, my first step was to fit isochrones to derive stellar masses from photometry. For this, I used isochrones from the Padova database (Marigo et al. 2017). The age, distance modulus  $(m - M)_0$ , metallicity  $[M/H]$ , and reddening  $E(B - V)$  were taken from Brogaard et al. (2017) for 47 Tuc, and from Milone et al. (2023b) for BS 90 and NGC 1783 (see Tab. 3.1 for details).

|                 | Age [Gyr] | $(m - M)_0$ [mag] | $E(B - V)$ [mag] | [M/H] [dex] |
|-----------------|-----------|-------------------|------------------|-------------|
| <b>47 Tuc</b>   | 11.8      | 13.21             | 0.02             | -0.45       |
| <b>BS 90</b>    | 4.2       | 18.91             | 0.03             | -0.7        |
| <b>NGC 1793</b> | 1.95      | 18.51             | 0.03             | -0.4        |

**Table 3.1:** Parameters used for the isochrones fitting. These values were derived from Brogaard et al. (2017) for 47 Tuc and from Milone et al. (2023b) for BS 90 and NGC 1793.



**Figure 3.13:** Isochrone fitting and mass ratio determination for 47 Tuc. The blue line is the Padova isochrone, the star-shaped markers refers to the members of the binary system that is the closest to the HB (on the right) and the RGB (on the left) but is still distinguishable from them. In the left panel the procedure is applied to the HB while in the right panel is applied to the RGB.

## Mass Ratio Determination

Once the isochrones were fitted, I used the synthetic binary systems that I had used as a boundary between region A and region B in the quantitative analysis to identify the members of the binary systems that are close to the HB (and later the RGB) while still being distinguishable from them. I then retrieved the initial masses of these stars through the isochrones, based on their magnitudes in the F814W filter. Using these masses, I derived the mass ratio  $q = \mathcal{M}_2/\mathcal{M}_1$ .

An example for this procedure applied to 47 Tuc can be seen in Fig. 3.13. The outcome for BS 90 and NGC 1783 is shown in Fig. A.6 and Fig. A.7 in the Appendix. In Tab. 3.2 I summarize all the results, the derived binary fractions and the relative mass ratios.

As a first result, I compared the binary fraction in the MS with the values I obtained for the HB and RGB. In 47 Tuc, Milone et al. (2012) found  $f_{bin}^{q>0.5} = 0.005 \pm 0.003$ , which is significantly lower than my results for the HB ( $f_{bin}^{q>0.9} = 0.040 \pm 0.012$ ) and RGB

|                 | HB                  |                   |           | RGB               |           |
|-----------------|---------------------|-------------------|-----------|-------------------|-----------|
|                 | $f_{bin}^{optical}$ | $f_{bin}^{UV}$    | $q_{min}$ | $f_{bin}$         | $q_{min}$ |
| <b>47 Tuc</b>   | $0.040 \pm 0.012$   | $0.040 \pm 0.012$ | 0.993     | $0.020 \pm 0.004$ | 0.904     |
| <b>BS 90</b>    | $0.046 \pm 0.024$   | $0.048 \pm 0.035$ | 0.996     | $0.038 \pm 0.022$ | 0.822     |
| <b>NGC 1783</b> | $0.093 \pm 0.021$   | $0.094 \pm 0.020$ | 0.816     | -                 | -         |

**Table 3.2:** Summary of the results. The table is divided between the results for the HB and the ones for the RGB. For the HB the binary fractions are shown for both the optical and the UV filters while for the RGB only the optical one is shown.

( $f_{bin}^{q>0.9} = 0.020 \pm 0.004$ ). In contrast, for NGC 1783, Mohandas et al. (2024) reported  $f_{bin}^{q>0.7} = 0.094 \pm 0.007$ , which is consistent with my result for the HB ( $f_{bin}^{q>0.8} = 0.093 \pm 0.020$ ) except for the lower limit for  $q$ . For BS 90, a direct comparison was not possible, as the binary fraction for the MS has not yet been determined for this cluster.

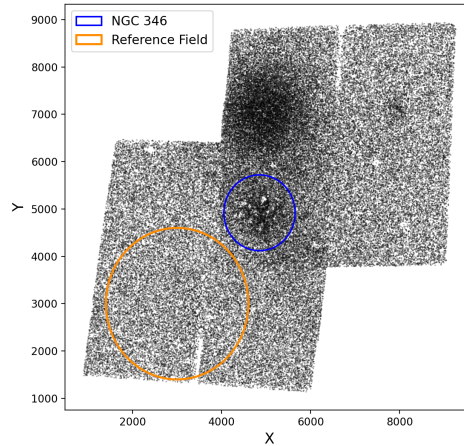
### 3.5 Total Fraction of Binaries

The procedure described in the previous sections allowed me to directly measure the fraction of binaries with  $q$  greater than the values shown in Tab. 3.2 without making any assumption about the mass ratio distribution  $f(q)$ . However, binaries with small mass ratios are indistinguishable from single stars therefore any attempt to determine the total fraction of binaries without assumptions on the mass-ratio distribution is impossible with this approach. To be consistent with previous studies (Milone et al. 2012), I adopted a flat  $f(q)$ . i.e. I assumed a constant mass-ratio distribution for all  $q$  values.

If I extrapolate the total binary fraction assuming a flat  $f(q)$ , the resulting values are extremely high and likely unreliable. For instance, following the same approach as Milone et al. (2012) and extrapolating the total binary fraction for 47 Tuc from my result,  $f_{bin}^{q>0.99} = 0.040 \pm 0.012$ , would yield a fraction exceeding 100%, which is clearly unphysical. Going in the details, to extrapolate the total binary fraction assuming a flat mass ratio distribution  $f(q)$ , I need to integrate  $f(q)$  over the detectable range of  $q$  and scale it to the full range  $0 < q < 1$ . From my result for 47 Tuc I have that 4% of stars are in binaries with  $q > 0.99$ . If I assume a uniform  $f(q)$ , the total binary fraction can be estimated as:

$$f_{bin}^{total} = f_{bin}^{q>q_{min}} \times \frac{1}{1 - q_{min}} \quad (3.5)$$

where  $q_{min} = 0.99$  in this case. Substituting the values the results would be an absurd binary fraction of 400%, suggesting that a flat  $f(q)$  distribution results in a paradox.



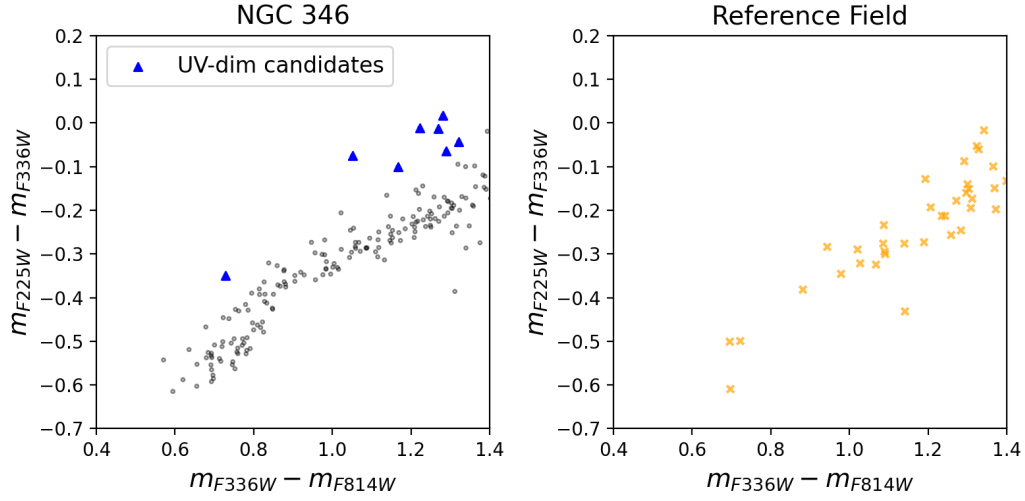
**Figure 3.14:** NGC 346 and the reference field used to account for field contamination. The area of the reference field is the twice as large as the one for NGC 346.

It is important to highlight that this study lacks an artificial star test. However, given the exceptional quality of the data used, the best currently available, I do not anticipate significant differences when incorporating it in future analyses. These results and subsequent implications will be discussed in the next chapter.

### 3.6 UV-dim stars in NGC 346

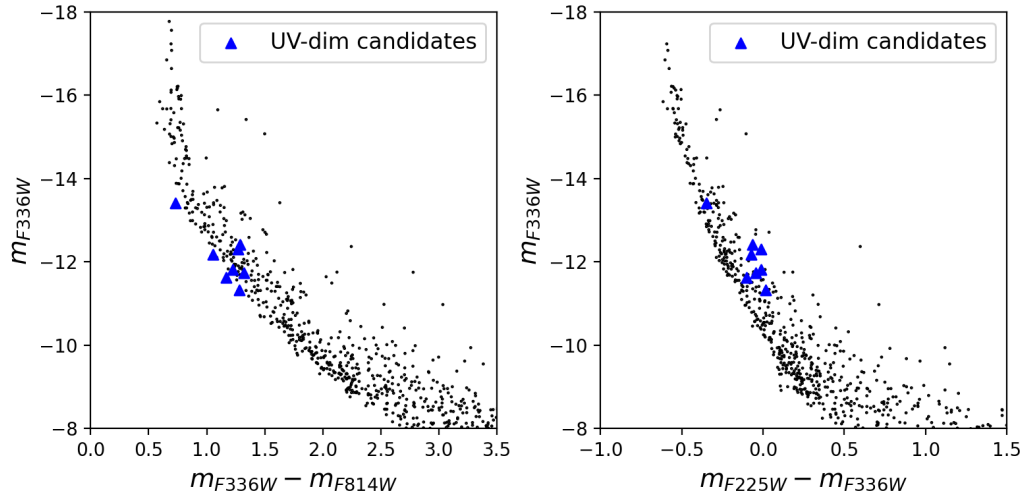
As already introduced in Chapter 1, UV-dim stars are stars extremely dim in ultraviolet bands and therefore located on the red side of the eMSTO in CMDs composed of UV filters. These stars were discovered in the cluster NGC 1783 (Milone et al. 2023b), but as investigated in Milone et al. (2023a), in which they analyzed five clusters in the Magellanic Clouds, they are a common feature of young clusters. As their nature is still debated, a way to shed light on this problem is to investigate their presence in a very young, star-forming cluster, to see whether these stars are present from the formation of the cluster or if they develop later.

Therefore, I investigated the presence of UV-dim stars in NGC 346, a  $\sim 3$  Myr cluster in the SMC, located south of BS 90. Since NGC 346 lies within the same field of view as BS 90, for which I already performed the data reduction, it provides an ideal target, allowing me to avoid redundant data processing. To account for field contamination, I defined the reference field as a circular region in the outskirts of NGC 346, with an area twice as large as the one used for NGC 346 itself (Fig. 3.14).



**Figure 3.15:** On the left  $m_{F225W} - m_{F336W}$  vs  $m_{F336W} - m_{F814W}$  two color diagram for NGC 346, the blue triangles are the UV-dim candidates. On the right the same two-color diagram.

To search for UV-dim stars in this cluster, I used the two-color diagram  $m_{F225W} - m_{F336W}$  vs  $m_{F336W} - m_{F814W}$ , which is an effective tool for identifying UV-dim stars (Milone et al. 2023a). I selected the UV-dim candidates by first defining a fiducial line for the narrow sequence of stars in this diagram, then selecting stars with a  $m_{F225W} - m_{F336W}$  color at least 0.1 magnitudes greater than the fiducial line. This process led to the identification of 8 UV-dim candidates. To estimate the contamination from field stars in the two-color diagram of NGC 346 I analyzed the same diagram for stars in the reference field. The results are illustrated in Fig. 3.15, while in Fig. 3.16 I present two CMDs, namely  $m_{F336W}$  vs  $m_{F336W} - m_{F814W}$  and  $m_{F336W}$  vs  $m_{F225W} - m_{F336W}$  to investigate where the candidate UV-dim stars are located.. The lack of reference-field stars in the portion of the two-color diagram populated by candidate UV-dim stars indicates negligible contamination from field stars.



**Figure 3.16:**  $m_{F336W}$  vs  $m_{F336W} - m_{F814W}$  and  $m_{F336W}$  vs  $m_{F225W} - m_{F336W}$  CMDs for NGC 346. The blue triangles are the candidate UV-dim stars.



## Chapter 4

# Discussion and Conclusions

Binary stars are among the major driving forces of star cluster dynamical evolution, affecting their structural parameters, mass function, and long-term stability. Even though MS binaries have been very well researched, their presence and properties within evolved stellar populations are not so well known. In this thesis, I explored the phenomenon of binary fractions in the HB and RGB in three clusters with diverse ages. Using a method of binary detection appropriate to these evolutionary stages, I reported new findings on the existence and detectability of binaries outside the MS. I also examined the presence of UV-dim stars in the very young cluster NGC 346, contributing to the debate regarding their nature and evolutionary implications. The presence of UV-dim candidates in such a young cluster is a strong indication that this is an inherent aspect of early cluster evolution and may contradict theory as it stands.

In the following sections, I summarize the principal findings of this research, discuss their broader implications, and propose possible directions for future investigation.

### 4.1 Binaries beyond the MS

As demonstrated in the previous chapter, the binary fractions derived for both the HB and RGB in all three clusters differ from those reported in the literature for the MS. For instance, in the case of 47 Tucanae, the binary fraction for the HB was found to be  $f_{bin}^{q>0.9} = 0.040 \pm 0.012$ , whereas the corresponding value for the MS is significantly lower at  $f_{bin}^{q>0.5} = 0.009 \pm 0.003$  (Milone et al. 2012). In contrast, for NGC 1783, the results are more consistent, with my analysis yielding for the HB  $f_{bin}^{q>0.8} = 0.093 \pm 0.021$ , compatible with the MS fraction of  $f_{bin}^{q>0.7} = 0.094 \pm 0.007$  found by Mohandasan et al. (2024).

Extrapolating the total binary fraction from HB and RGB fractions, under the assumption of a flat mass-ratio distribution, leads to a paradox: the total binary fraction that exceeds the total number of stars in the cluster. This discrepancy suggests two possible explanations.

The first is that the mass distribution is peaked around one and does not conform to the assumption of a flat distribution, as previously postulated by Milone and colleagues in 2012 and subsequent works. It should be noted that the flatness of mass-ratio distribution is a statistical conclusion drawn on the basis of the 59 clusters that have been studied in Milone et al. (2012) and does not rule out deviations from this pattern. It is also noteworthy that the analysis used in that study was restricted to the outermost regions of the clusters. Consequently, the mass-ratio distribution within the cluster’s interior might deviate from the observed flatness due to the effects of mass segregation. In order to ensure greater consistency with the results found by Milone et al. (2012) for 47 Tucanae, I repeated the entire analysis excluding the cluster center. This was done to check whether the results would be different from the previous conclusion. However, the results for both the binary fraction and mass-ratio remained consistent even after excluding the center. Consequently, it can be inferred that the mass-ratio distribution should be peaked around one, as found by Fisher et al. (2005), which demonstrated a distinct peak in the  $q$  distribution of field binaries near unity. This hypothesis makes the binary fractions inferred in the HB and the RGB compatible with the ones found in literature for the MS.

The second explanation is that the presence of high binary fractions may be caused by evolved Blue Stragglers stars (BSSs) within the sample regions, both in the RGB and the HB. BSSs are MS stars that appear brighter and bluer than the main-sequence turn-off stars, making them unexpected in a CMD. When isochrones are fitted to the CMD, these stars are found to be consistent with a younger stellar population. One of the proposed formation mechanisms for BSSs suggests that they originate from binary systems undergoing mass transfer. Given that BSSs resemble a younger population, their evolved counterparts would be located to the left of the RGB (in a bluer region of the CMD) and in the upper region of the HB, above the bulk of HB stars (Ferraro et al. 2016). This is precisely where the regions used to compute the RGB and HB binary fraction were selected. This suggests a potential method to estimate the fraction of BSSs by comparing the binary fractions in the MS respectively with the ones found in the RGB and the HB, under the assumption of a flat mass ratio distribution. Following this hypothesis I computed the BSSs fraction in the RGB and HB for 47 Tuc and the BSSs fraction in the HB for NGC 1783, while for BS 90

this analysis was not possible as currently there are no studies about the binary fraction in the MS for this cluster.

### Blue Stragglers Fraction

To find the BSSs fraction in 47 Tucanae, I first need to derive the binary fraction in the main sequence with  $q > 0.9$ . Starting from the fraction reported in Milone et al. (2012) of  $f_{bin}^{q>0.5} = 0.009 \pm 0.003$  and assuming a flat mass-ratio distribution, the fraction of binaries with  $q > 0.9$  is  $f_{bin} = 0.0018 \pm 0.0006$ . The binary fraction in the RGB is  $f_{bin}^{q>0.9} = 0.020 \pm 0.004$ , so the fraction of BSSs can be found by subtracting those two numbers. For 47 Tuc, I found a Blue Stragglers fraction in the RGB of  $f_{BSS} = 0.018 \pm 0.004$ . I then applied the same method to the HB, finding an evolved BSSs fraction of  $f_{BSS} = 0.032 \pm 0.012$ . I repeated this calculation for NGC 1783, where Mohandasan et al. (2024) found a binary fraction in the MS of  $f_{bin}^{q>0.7} = 0.094 \pm 0.007$  (from which  $f_{bin}^{q>0.8} = 0.063 \pm 0.005$ ), I found an evolved BSSs fraction in the HB for this cluster of  $f_{BSS} = 0.030 \pm 0.022$ . It is important to note that these fractions should be considered as upper limits for the BSSs population in these clusters.

The efficacy of this method suggests the potential for extending binary fraction studies to a more extensive range of clusters. This includes Galactic clusters, where MS binary analysis is challenging due to crowding, and extragalactic clusters, such as those in the Andromeda Galaxy, where the available photometry does not reach the MS. Furthermore, this method can be used to determine the fraction of Blue Stragglers.

## 4.2 The enigma of UV-dim stars

The identification of candidate UV-dim stars in the  $\sim 3$  Myr cluster NGC 346 is a novel result that has the potential to invalidate theories about the nature of these peculiar stars, as they may have existed since the cluster formation. The current hypothesis regarding the nature of such stars is based on them being obscured by a dust ring, which reduces their flux in UV bands (D’Antona et al. 2023). However, D’Antona and colleagues suggest that UV-dim stars are fast rotators and establish a link between fast rotation and mass loss events, which can lead to the formation of circumstellar dust. Yet, the placement of UV-dim stars on a CMD shows them occupying the blue MS, an area inhabited by slow rotators. This means that the low UV flux can be accounted for by the presence of a circumstellar dust ring, but the nature of this ring is still to be fully determined.

The presence of UV-dim stars in a cluster so young provides strong argument against the hypothesis that they were once rapid rotators and then underwent a transition to slow rotators after being subject to some form of braking mechanism. While this hypothesis was possible for clusters of  $\sim 200$  Myr, it is not applicable to a cluster which is a few million years old. Another way to test for the presence of a circumstellar disk would be to check whether the flux of these stars varies over time. A variation in flux would assist in highlighting dishomogeneities, thereby determining the presence of a circumstellar disk. Unfortunately, the data set available to me does not allow me to do so.

### 4.3 Final Remarks

This work significantly expands the investigation of binary fractions beyond the MS, providing novel information on the incidence and nature of binary stars in the HB and RGB of star clusters, as well as a potential method to determine the fraction of Blue Stragglers. The detection of UV-dim candidates in a very young cluster contradicts existing formation scenarios and raises new questions about their nature.

These results offer valuable contributions and highlights areas for future research. The next step is to extend this method to a larger sample of clusters with varying masses and metallicities, is the next step. This will establish if the trends seen here are universal, allowing us to extend the binary analysis to extragalactic clusters. Artificial star tests will refine binary fraction determinations, while spectroscopy of UV-dim stars will help characterize the composition and geometry of their circumstellar dust rings. These steps are necessary to further our knowledge of binaries and to comprehend the nature of UV-dim stars, which will ultimately improve our understanding of stellar populations and their evolution.

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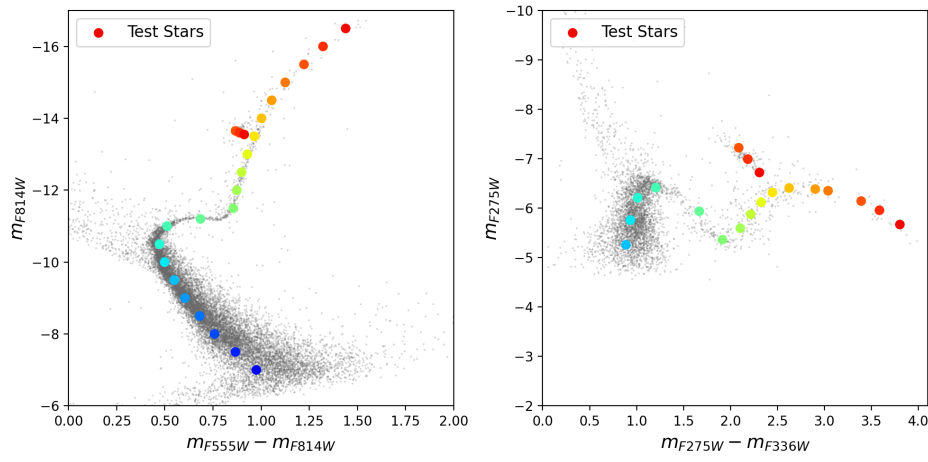
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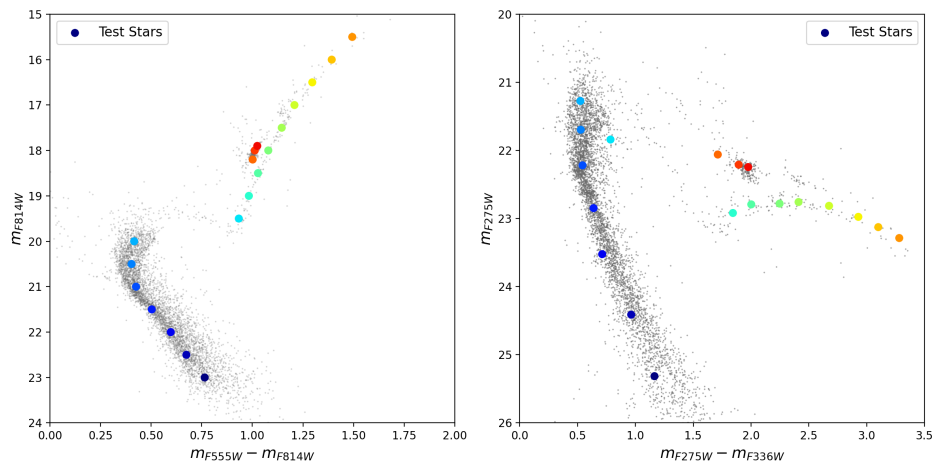


# Appendix A

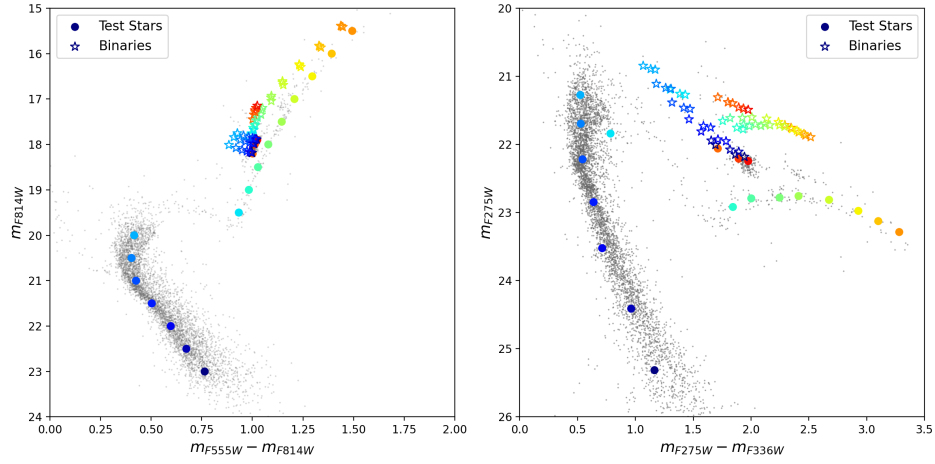
## Additional Figures



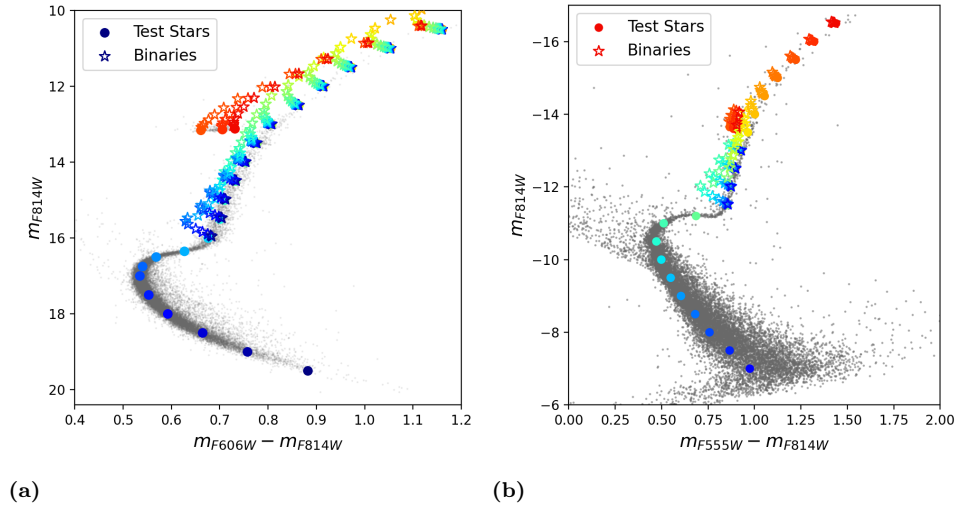
**Figure A.1:** Optical and UV CMDs for BS 90, displaying the test stars as filled circles.



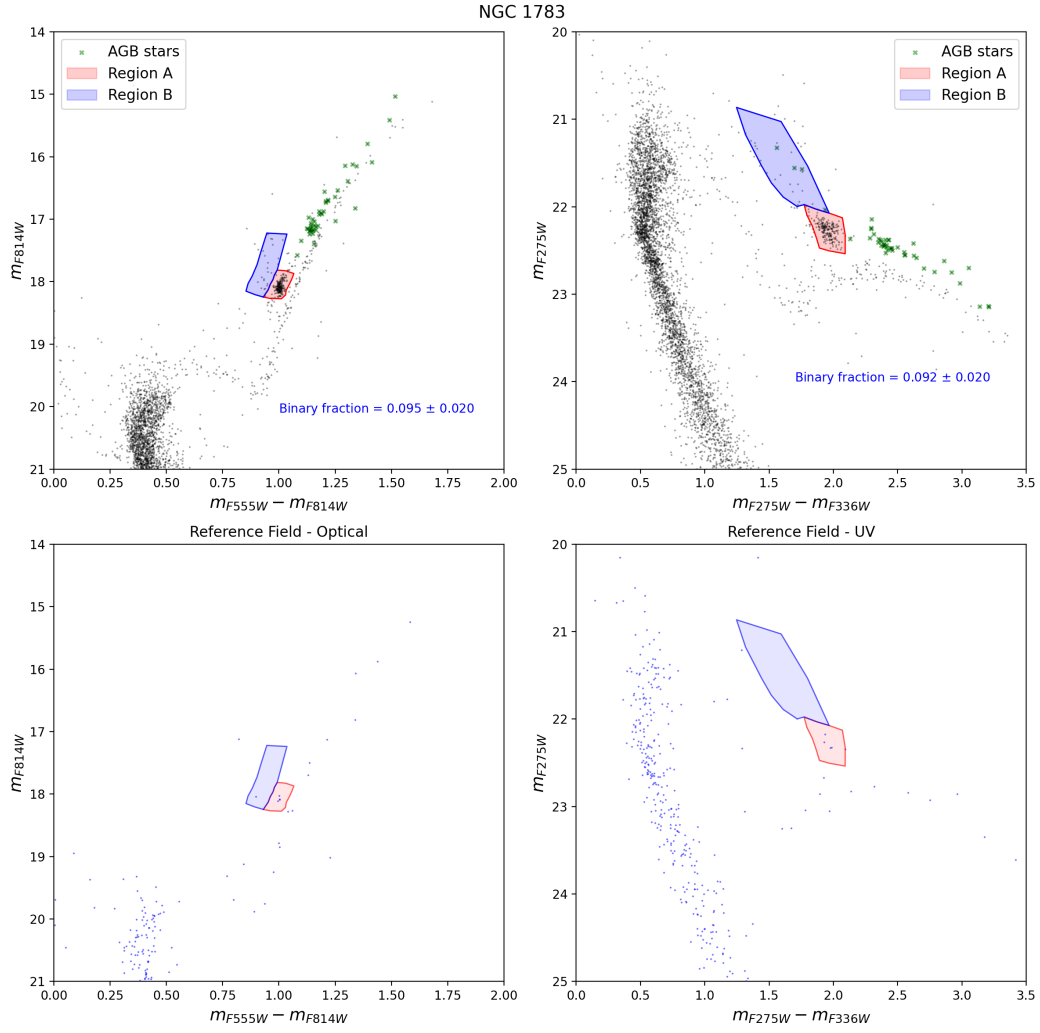
**Figure A.2:** Optical and UV CMDs for NGC 1783, displaying the test stars as filled circles.

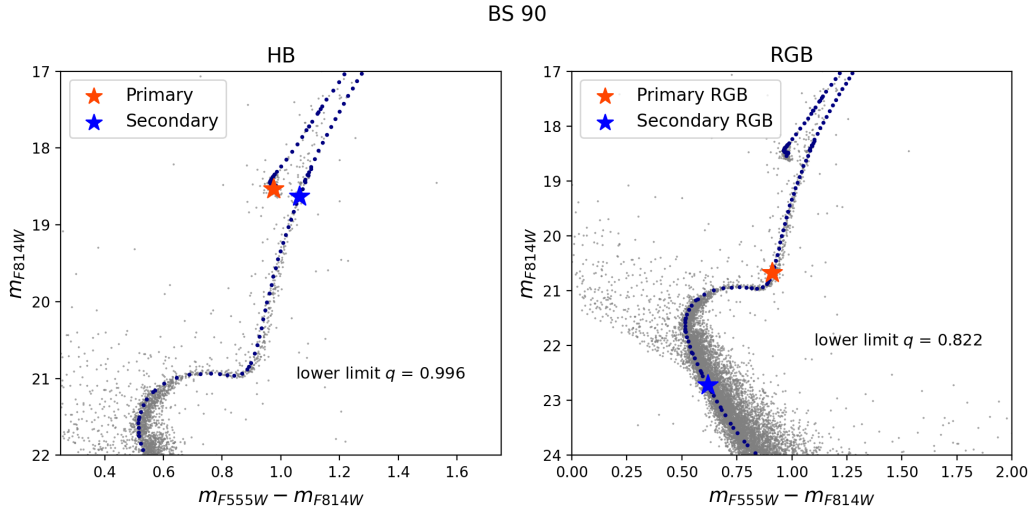


**Figure A.3:** Optical and UV CMDs for NGC 1783, displaying the test stars as filled circles and the synthetic binaries with empty star-shaped markers.

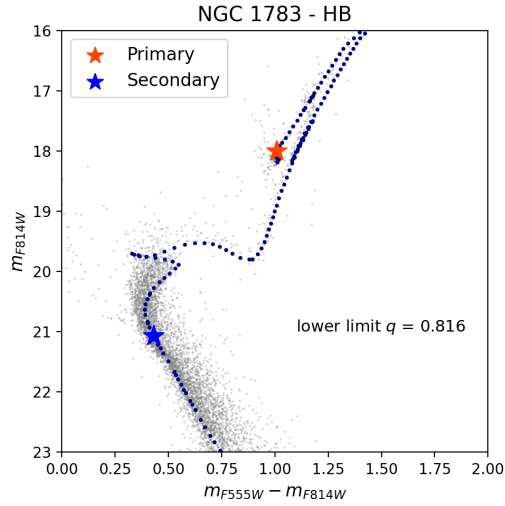


**Figure A.4:** Optical CMDs for 47 Tuc (left panel) and for BS 90 (right panel), displaying the test stars as filled circles and the synthetic binaries with empty star-shaped markers. The binary systems are constructed using all RGB stars as primaries, compromising the readability of the CMDs but offering a comprehensive overview of the distribution of these binary systems.





**Figure A.6:** Isochrone fitting and mass ratio determination for BS 90. The blue line is the Padova isochrone, the star-shaped markers refers to the members of the binary system that is the closest to the HB (on the right) and the RGB (on the left) but is still distinguishable from them. In the left panel the procedure is applied to the HB while in the right panel is applied to the RGB.



**Figure A.7:** Isochrone fitting and mass ratio determination for NGC 1783 for binaries in the HB. The blue line is the Padova isochrone, the star-shaped markers refers to the members of the binary system that is the closest to the HB but is still distinguishable from it.

## Appendix B

### Complementary Data

| $m_{F814W}$ | $m_{F606W}$ | $m_{F435W}$ | $m_{F336W}$ | $m_{F275W}$ |
|-------------|-------------|-------------|-------------|-------------|
| 19.5000     | 20.3826     | 21.7230     | 22.5073     | 25.0779     |
| 19.0000     | 19.7576     | 20.8966     | 21.3260     | 23.5535     |
| 18.5000     | 19.1648     | 20.1414     | 20.2540     | 21.9633     |
| 18.0000     | 18.5927     | 19.4358     | 19.3551     | 20.7060     |
| 17.5000     | 18.0533     | 18.8103     | 18.6448     | 19.7132     |
| 17.0000     | 17.5349     | 18.2610     | 18.0684     | 19.0700     |
| 16.7500     | 17.2905     | 18.0209     | 17.8631     | 18.8654     |
| 16.5000     | 17.0687     | 17.8663     | 17.7906     | 18.8633     |
| 16.3500     | 16.9772     | 17.8608     | 17.9633     | 19.2771     |
| 16.0000     | 16.6764     | 17.6788     | 17.9198     | 19.5588     |
| 15.5000     | 16.2005     | 17.2369     | 17.5239     | 19.2602     |
| 15.0000     | 15.7099     | 16.7797     | 17.1372     | 18.9023     |
| 14.5000     | 15.2327     | 16.3343     | 16.7795     | 18.6314     |
| 14.0000     | 14.7533     | 15.9051     | 16.4302     | 18.4515     |
| 13.5000     | 14.2766     | 15.4969     | 16.1349     | 18.3018     |
| 13.0000     | 13.8076     | 15.0902     | 15.8800     | 18.2103     |
| 12.5000     | 13.3640     | 14.7510     | 15.7359     | 18.4206     |
| 12.0000     | 12.9152     | 14.4318     | 15.6264     | 18.5014     |
| 11.5000     | 12.4725     | 14.0987     | 15.5233     | 18.6595     |
| 11.0000     | 12.0538     | 13.8674     | 15.5896     | 18.7700     |
| 10.5000     | 11.6604     | 13.6240     | 15.7225     | 18.8621     |
| 10.0000     | 11.3560     | 13.4752     | 15.8408     | 18.9802     |
| 13.1600     | 13.8212     | 14.7671     | 15.1880     | 16.6050     |
| 13.1400     | 13.8457     | 14.8260     | 15.2800     | 16.9420     |
| 13.1200     | 13.8517     | 14.9268     | 15.3820     | 17.2590     |

**Table B.1:** Magnitudes of the test stars used to build synthetic binaries for 47 Tucanae.

## Complementary Data

| $m_{F814W}$ | $m_{F555W}$ | $m_{F475W}$ | $m_{F336W}$ | $m_{F275W}$ |
|-------------|-------------|-------------|-------------|-------------|
| -16.5000    | -15.0608    | -14.3444    | -9.4722     | -5.6693     |
| -16.0000    | -14.6785    | -14.0123    | -9.5435     | -5.9578     |
| -15.5000    | -14.2773    | -13.6455    | -9.5356     | -6.1446     |
| -15.0000    | -13.8747    | -13.2424    | -9.3918     | -6.3531     |
| -14.5000    | -13.4450    | -12.8305    | -9.2880     | -6.3859     |
| -14.0000    | -12.9977    | -12.4060    | -9.0328     | -6.4086     |
| -13.5000    | -12.5350    | -11.9574    | -8.7646     | -6.3182     |
| -13.0000    | -12.0716    | -11.5182    | -8.4482     | -6.1215     |
| -12.5000    | -11.6014    | -11.0763    | -8.0912     | -5.8759     |
| -12.0000    | -11.1263    | -10.6234    | -7.6990     | -5.5951     |
| -11.5000    | -10.6445    | -10.1587    | -7.2778     | -5.3632     |
| -11.2000    | -10.5145    | -10.0982    | -7.6062     | -5.9382     |
| -11.0000    | -10.4886    | -10.1364    | -7.6213     | -6.4169     |
| -10.5000    | -10.0280    | -9.7251     | -7.2224     | -6.2109     |
| -10.0000    | -9.5014     | -9.1959     | -6.6928     | -5.7591     |
| -9.5000     | -8.9511     | -8.6352     | -6.1458     | -5.2591     |
| -9.0000     | -8.3957     | -8.0516     | -5.5791     | -4.7591     |
| -8.5000     | -7.8190     | -7.4241     | -5.0000     | -4.2591     |
| -8.0000     | -7.2419     | -6.7771     | -4.4210     | -3.7591     |
| -7.5000     | -6.6335     | -6.1115     | -3.8420     | -3.2591     |
| -7.0000     | -6.0248     | -5.4581     | -3.2630     | -2.7591     |
| -13.6500    | -12.7831    | -12.2770    | -9.3105     | -7.2228     |
| -13.6000    | -12.7127    | -12.2005    | -9.1762     | -6.9922     |
| -13.5500    | -12.6380    | -12.1168    | -9.0293     | -6.7215     |

**Table B.2:** Instrumental magnitudes of the test stars used to build synthetic binaries for BS 90.

| $m_{F814W}$ | $m_{F555W}$ | $m_{F435W}$ | $m_{F343W}$ | $m_{F336W}$ | $m_{F275W}$ |
|-------------|-------------|-------------|-------------|-------------|-------------|
| 23.0000     | 23.7645     | 24.2970     | 24.0353     | 24.1485     | 25.3138     |
| 22.5000     | 23.1739     | 23.6445     | 23.3699     | 23.4468     | 24.4108     |
| 22.0000     | 22.5968     | 23.0000     | 22.7329     | 22.8095     | 23.5226     |
| 21.5000     | 22.0030     | 22.3533     | 22.1612     | 22.2076     | 22.8464     |
| 21.0000     | 21.4252     | 21.7097     | 21.6462     | 21.6750     | 22.2183     |
| 20.5000     | 20.9032     | 21.1522     | 21.1489     | 21.1678     | 21.6957     |
| 20.0000     | 20.4161     | 20.6806     | 20.6774     | 20.7493     | 21.2737     |
| 19.5000     | 20.4331     | 21.1065     | 20.3699     | 21.0548     | 21.8398     |
| 19.0000     | 19.9828     | 20.7772     | 20.9146     | 21.0747     | 22.9180     |
| 18.5000     | 19.5277     | 20.3729     | 20.6484     | 20.7897     | 22.7918     |
| 18.0000     | 19.0787     | 19.9806     | 20.3796     | 20.5333     | 22.7803     |
| 17.5000     | 18.6452     | 19.6000     | 20.1953     | 20.3462     | 22.7572     |
| 17.0000     | 18.2080     | 19.2445     | 20.0505     | 20.1384     | 22.8129     |
| 16.5000     | 17.7961     | 18.9155     | 19.9000     | 20.0468     | 22.9748     |
| 16.0000     | 17.3922     | 18.6099     | 19.8138     | 20.0252     | 23.1255     |
| 15.5000     | 16.9935     | 18.3194     | 19.7331     | 20.0043     | 23.2859     |
| 18.2000     | 19.2016     | 19.9902     | 20.1679     | 20.3470     | 22.0598     |
| 18.0000     | 19.0120     | 19.8581     | 20.1489     | 20.3157     | 22.2091     |
| 17.9000     | 18.9243     | 19.7852     | 20.1174     | 20.2684     | 22.2440     |

**Table B.3:** Magnitudes of the test stars used to build synthetic binaries for NGC 1783.